# calculus-ai999x: capability guide

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Every capability calculus-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

## Quick start

```
import calculus_ai999x

result = calculus_ai999x.solve('derivative_at_point', f='x^3 - 2x', a=2)
result.show()
# derivative = 10, derivative_expression = 3*x^2 - 2
```

## By topic

| topic | capabilities |
|---|---|
| Derivatives | Derivative at a point, Second derivative at a point, Higher derivative at a point, Implicit differentiation, Mean value theorem |
| Derivatives / concavity | Inflection points |
| Derivatives / curvature | Curvature of a graph |
| Derivatives / linearization | Linear approximation |
| Derivatives / motion | Velocity and acceleration from position |
| Derivatives / optimization | Absolute extrema on a closed interval, Critical points and their classification |
| Derivatives / rates of change | Average rate of change |
| Derivatives / tangent and normal lines | Tangent line, Normal line |
| Fourier series | Fourier coefficients |
| Integrals | Definite integral, Derivative of an integral (fundamental theorem) |
| Integrals / applications | Average value of a function |
| Integrals / arc length | Arc length of a graph |
| Integrals / area | Area between two curves, Area enclosed by two curves |
| Integrals / centroids | Centroid of a region under a curve |
| Integrals / improper integrals | Improper integral to infinity |
| Integrals / Riemann sums | Left Riemann sum, Right Riemann sum |
| Integrals / surface area | Surface area of revolution |

| topic | capabilities |
|---|---|
| Integrals / volumes of revolution | Volume of revolution (disk method), Volume of revolution (washer method), Volume of revolution (shell method) |
| Limits | Limit at a point, Limit at infinity |
| Multiple integrals | Double integral over a rectangle, Double integral over a region, Triple integral over a box, Surface area of a graph |
| Multivariable calculus | Partial derivative, Gradient, Directional derivative, Tangent plane |
| Multivariable calculus / change of variables | Jacobian determinant |
| Multivariable calculus / optimization | Second derivative test (two variables) |
| Numerical differentiation | Finite-difference derivative |
| Numerical integration | Trapezoidal rule, Simpson's rule, Midpoint rule, Gauss-Legendre quadrature |
| Numerical methods / root finding | Newton's method |
| Parametric curves | Arc length of a parametric curve, Slope and concavity of a parametric curve |
| Polar curves | Area in polar coordinates |
| Series | Sum of an infinite geometric series, Partial sum of a geometric series, Sum of an arithmetic series, Sum of an infinite series, Ratio test |
| Series / Taylor | Taylor polynomial |
| Special functions | Gamma function, Beta function, Error function, Regularized incomplete gamma functions, Bessel function of the first kind |
| Special functions / orthogonal polynomials | Legendre polynomial, Chebyshev polynomial |
| Special functions / series | Riemann zeta function |
| Vector calculus | Laplacian, Divergence, Curl, Line integral of a vector field, Line integral with respect to arc length, Green's theorem on a rectangle, Outward flux through a box (divergence theorem) |

## Outcomes

| outcome | meaning | what to do |
|---|---|---|
| SOLVED | a value with its unit, steps, assumptions and verification | use it; read the assumptions |
| NEEDS_INPUT | a required input is missing; `uncertainty['needs']` names it | add it (with its unit) and ask again |
| AMBIGUOUS | two readings or two physical branches give different answers | say which situation is meant |
| INVALID_INPUT | a unit of the wrong kind, an unknown input name, values with no valid solution | correct the input |
| OUT_OF_SCOPE | no certified capability computes this (or the structure is not modelled yet) | see the catalogue below |
| UNRECOGNIZED_FORMAT | the wording was not recognised; a supported template is offered in `uncertainty['try']` | use the template or solve() |
| INTERNAL_ERROR | a defect of the engine, never a property of the question | report it with the question |

## Result fields

| field | meaning |
|---|---|
| value, unit | the result in SI units |
| values | every output of a procedure (x1, x2, discriminant ...) |
| outcome | one of the seven outcomes above |
| steps | each equation used, its inputs (with where each came from) and its result |
| assumptions | conditions and declared defaults the answer relies on |
| verification | residual of every equation, dimensional consistency, independent checks |
| latex() | the derivation as LaTeX |
| show() | renders in a notebook, prints in a terminal |

## Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

## Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

| id | operation | inputs | outputs |
|---|---|---|---|
| derivative_at_point | Derivative at a point | f, a | derivative, derivative_expression |
| second_derivative_at_point | Second derivative at a point | f, a | second_derivative, second_derivative_expression |
| nth_derivative_at_point | Higher derivative at a point | f, n, a | nth_derivative, nth_derivative_expression |
| tangent_line | Tangent line | f, a | slope, intercept, y0, equation |
| normal_line | Normal line | f, a | slope, intercept, y0 |
| average_rate_of_change | Average rate of change | f, a, b | average_rate, fa, fb |
| velocity_acceleration | Velocity and acceleration from position | s, t0 | velocity, acceleration, speed, position |
| linear_approximation | Linear approximation | f, a, x1 | approximation, actual, error, slope |
| curvature | Curvature of a graph | f, a | curvature, radius_of_curvature |
| definite_integral | Definite integral | f, a, b | integral |
| average_value | Average value of a function | f, a, b | average_value, integral |
| area_between_curves | Area between two curves | f, g, a, b | area, net_integral, crossings |
| area_enclosed_curves | Area enclosed by two curves | f, g | area, x_left, x_right, intersections |

| id | operation | inputs | outputs |
|---|---|---|---|
| disk_volume | Volume of revolution (disk method) | f, a, b | volume |
| washer_volume | Volume of revolution (washer method) | f, g, a, b | volume |
| shell_volume | Volume of revolution (shell method) | f, a, b | volume |
| arc_length | Arc length of a graph | f, a, b | arc_length |
| surface_area_revolution | Surface area of revolution | f, a, b | surface_area |
| centroid_region | Centroid of a region under a curve | f, a, b | area, x_bar, y_bar |
| improper_integral | Improper integral to infinity | f, a | integral |
| left_riemann_sum | Left Riemann sum | f, a, b, n | approximation |
| right_riemann_sum | Right Riemann sum | f, a, b, n | approximation |
| trapezoid_rule | Trapezoidal rule | f, a, b, n | approximation, exact, error |
| simpson_rule | Simpson's rule | f, a, b, n | approximation, exact, error |
| midpoint_rule | Midpoint rule | f, a, b, n | approximation, exact, error |
| gauss_legendre | Gauss-Legendre quadrature | f, a, b, n | approximation, exact, error |
| numerical_derivative | Finite-difference derivative | f, a, h, method | approximation, exact, error |
| newton_method | Newton's method | f, x0, n | x_n, f_x_n, iterates |
| limit_at_point | Limit at a point | f, a | limit |
| limit_at_infinity | Limit at infinity | f | limit |
| implicit_derivative | Implicit differentiation | F, p | dy_dx |
| extrema_on_interval | Absolute extrema on a closed interval | f, a, b | absolute_max, absolute_min, critical_points |
| critical_points_classify | Critical points and their classification | f | critical_points, local_maxima, local_minima |
| inflection_points | Inflection points | f | inflection_points |
| mean_value_theorem | Mean value theorem | f, a, b | secant_slope, c_values |
| ftc_derivative | Derivative of an integral (fundamental theorem) | f, a, g, x0 | derivative |
| geometric_series_sum | Sum of an infinite geometric series | a, r | sum |
| geometric_partial_sum | Partial sum of a geometric series | a, r, n | sum |
| arithmetic_series_sum | Sum of an arithmetic series | a1, d, n | sum, last_term |
| series_sum | Sum of an infinite series | a, k | sum |
| ratio_test | Ratio test | a | ratio_limit, verdict |
| taylor_polynomial | Taylor polynomial | f, a, n, x1 | coefficients, value, function_value, error |
| partial_derivative | Partial derivative | f, var, p | partial |
| gradient | Gradient | f, p | gradient, magnitude |
| directional_derivative | Directional derivative | f, p, u | directional_derivative |
| tangent_plane | Tangent plane | f, p | z0, fx, fy, c |

| id | operation | inputs | outputs |
|---|---|---|---|
| hessian_classification | Second derivative test (two variables) | f, p | fxx, fxy, fyy, D, classification |
| laplacian | Laplacian | f, p | laplacian |
| divergence | Divergence | P, Q, R, p | divergence |
| curl | Curl | P, Q, R, p | curl |
| jacobian_determinant | Jacobian determinant | xu, yv, p | jacobian |
| double_integral_rectangle | Double integral over a rectangle | f, a, b, c, d | integral |
| double_integral_region | Double integral over a region | f, a, b, g1, g2 | integral |
| triple_integral_box | Triple integral over a box | f, x0, x1, y0, y1, z0, z1 | integral |
| surface_area_graph | Surface area of a graph | f, a, b, c, d | surface_area |
| line_integral_vector | Line integral of a vector field | P, Q, xt, yt, t0, t1 | line_integral |
| line_integral_scalar | Line integral with respect to arc length | f, xt, yt, t0, t1 | line_integral |
| greens_theorem | Green's theorem on a rectangle | P, Q, a, b, c, d | circulation, double_integral |
| divergence_theorem_box | Outward flux through a box (divergence theorem) | P, Q, R, x0, x1, y0, y1, z0, z1 | flux |
| parametric_arc_length | Arc length of a parametric curve | xt, yt, t0, t1 | arc_length |
| parametric_slope | Slope and concavity of a parametric curve | xt, yt, t0 | dy_dx, d2y_dx2 |
| polar_area | Area in polar coordinates | r, al, be | area |
| gamma_function | Gamma function | x | gamma |
| beta_function | Beta function | a, b | beta |
| error_function | Error function | x | erf, erfc |
| regularized_gamma | Regularized incomplete gamma functions | a, x | P, Q |
| bessel_j | Bessel function of the first kind | n, x | J |
| riemann_zeta | Riemann zeta function | s | zeta |
| legendre_polynomial | Legendre polynomial | n, x | P |
| chebyshev_polynomial | Chebyshev polynomial | n, x | T |
| fourier_coefficients | Fourier coefficients | f, L, k | a0, a_k, b_k |

### Derivative at a point — `derivative_at_point`

The derivative f'(a) of a function f(x) at $x = a$ (the slope of its graph there).

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `derivative` | f'(a) |
| `derivative_expression` | f'(x) as an expression |

- **Method:** exact symbolic differentiation (sum, product, quotient and chain rules), then evaluation at $x = a$
- **Assumes:** f is defined and differentiable at $x = a$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('derivative_at_point', f='x^3 - 2x', a=2)
# -> derivative = 10, derivative_expression = 3*x^2 - 2   (SOLVED)
```

```
calculus_ai999x.solve('derivative_at_point', f='sin(3x) + x^2', a=0)
# -> derivative = 3, derivative_expression = 2*x + 3*cos(3*x)   (SOLVED)
```

```
calculus_ai999x.solve('derivative_at_point', f='x^2*exp(x)', a=1)
# -> derivative = 8.15485, derivative_expression = x^2*exp(x) + 2*x*exp(x)   (SOLVED)
```

```
calculus_ai999x.solve('derivative_at_point', f='log(x)', a=4)
# -> derivative = 0.25, derivative_expression = 1/x   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the derivative of f(x) = x^3 - 2x at x = 2. | SOLVED | derivative = 10, derivative_expression = 3*x^2 - 2 |
| What is f'(0) if f(x) = sin(3x) + x^2? | SOLVED | derivative = 3, derivative_expression = 2*x + 3*cos(3*x) |
| Differentiate x^2*exp(x) and evaluate at x = 1. | SOLVED | derivative = 8.15485, derivative_expression = x^2*exp(x) + 2*x*exp(x) |
| What is the slope of the curve y = sqrt(x) at x = 9? | SOLVED | derivative = 0.166667, derivative_expression = 1/(2*sqrt(x)) |
| Find the instantaneous rate of change of x^2 + 3x at x = 1. | SOLVED | derivative = 5, derivative_expression = 2*x + 3 |
| Find the derivative of log(x) at x = -1. | INVALID_INPUT | Derivative at a point: the expression is not real there. |

## Second derivative at a point — `second_derivative_at_point`

The second derivative f''(a) of f(x) at $x = a$ (concavity: positive means concave up).

$$f''(a) = \frac{d^2 f}{dx^2}\Big|_{x=a}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `second_derivative` | f''(a) |
| `second_derivative_expression` | f''(x) |

- **Method:** exact symbolic differentiation twice, then evaluation at $x = a$
- **Assumes:** f is twice differentiable at $x = a$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('second_derivative_at_point', f='x^4 - 3x^2', a=1)
# -> second_derivative = 6, second_derivative_expression = 6*(2*x^2 - 1)   (SOLVED)
```

```
calculus_ai999x.solve('second_derivative_at_point', f='cos(2x)', a=0)
# -> second_derivative = -4, second_derivative_expression = -4*cos(2*x)   (SOLVED)
```

```
calculus_ai999x.solve('second_derivative_at_point', f='exp(2x)', a=0)
# -> second_derivative = 4, second_derivative_expression = 4*exp(2*x)   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the second derivative of x^4 - 3x^2 at x = 1. | SOLVED | second_derivative = 6, second_derivative_expression = 6*(2*x^2 - 1) |
| What is f''(0) if f(x) = cos(2x)? | SOLVED | second_derivative = -4, second_derivative_expression = -4*cos(2*x) |
| Find the second derivative of sqrt(x) at x = -4. | INVALID_INPUT | Second derivative at a point: the expression is not real there. |

## Higher derivative at a point — `nth_derivative_at_point`

The n-th derivative $f^{n}$(a) of f(x) at $x = a$.

$$f^{(n)}(a) = \frac{d^n f}{dx^n}\Big|_{x=a}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `n` | the order of the derivative | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `nth_derivative` | $f^{n}$(a) |
| `nth_derivative_expression` | $f^{n}$(x) |

- **Method:** exact symbolic differentiation n times, then evaluation at $x = a$
- **Assumes:** $1 \leq n \leq 12$; f is n times differentiable at $x = a$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('nth_derivative_at_point', f='sin(2x)', n=4, a=1)
# -> nth_derivative = 14.5488, nth_derivative_expression = 16*sin(2*x)   (SOLVED)
```

```
calculus_ai999x.solve('nth_derivative_at_point', f='x^5', n=3, a=2)
# -> nth_derivative = 240, nth_derivative_expression = 60*x^2   (SOLVED)
```

```
calculus_ai999x.solve('nth_derivative_at_point', f='exp(3x)', n=5, a=0)
# -> nth_derivative = 243, nth_derivative_expression = 243*exp(3*x)   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the 4th derivative of sin(2x) at x = 1. | SOLVED | nth_derivative = 14.5488, nth_derivative_expression = 16*sin(2*x) |
| Find the third derivative of x^5 at x = 2. | SOLVED | nth_derivative = 240, nth_derivative_expression = 60*x^2 |
| What is the 5th derivative of exp(3x) at x = 0? | SOLVED | nth_derivative = 243, nth_derivative_expression = 243*exp(3*x) |

### Tangent line — `tangent_line`

The tangent line $y = mx + b$ to $y = f$(x) at $x = a$: slope $m = f$'(a), intercept $b = f$(a) - a*f'(a).

$$y = f(a) + f'(a)\,(x - a)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point of tangency $x = a$ | |

| output | meaning |
|---|---|
| `slope` | slope $m = f$'(a) |
| `intercept` | y-intercept b |
| `y0` | f(a) |
| `equation` | the line as $y = mx + b$ |

- **Method:** point-slope form through (a, f(a)) with the exact derivative
- **Assumes:** f is differentiable at $x = a$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('tangent_line', f='x^2 + 1', a=2)
# -> slope = 4, intercept = -3, y0 = 5, equation = y = 4*x - 3   (SOLVED)
```

```
calculus_ai999x.solve('tangent_line', f='sin(x)', a=0)
# -> slope = 1, intercept = 0, y0 = 0, equation = y = 1*x + 0   (SOLVED)
```

```
calculus_ai999x.solve('tangent_line', f='exp(x)', a=1)
# -> slope = 2.71828, intercept = 0, y0 = 2.71828, equation = y = 2.71828182846*x + 0   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the equation of the tangent line to y = x^2 + 1 at x = 2. | SOLVED | slope = 4, intercept = -3, y0 = 5, equation = y = 4*x - 3 |
| Find the tangent line to f(x) = sin(x) at x = 0. | SOLVED | slope = 1, intercept = 0, y0 = 0, equation = y = 1*x + 0 |
| What is the slope of the tangent line to y = x^3 at x = -1? | SOLVED | slope = 3, intercept = 2, y0 = -1, equation = y = 3*x + 2 |

## Normal line — `normal_line`

The normal line (perpendicular to the tangent) to $y = f$(x) at $x = a$: slope -1/f'(a).

$$y = f(a) - \frac{1}{f'(a)}\,(x - a)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `slope` | slope -1/f'(a) |
| `intercept` | y-intercept |
| `y0` | f(a) |

- **Method:** perpendicular slope -1/m through (a, f(a))
- **Assumes:** f'$(a) \neq 0$ (otherwise the normal is vertical)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('normal_line', f='x^2', a=1)
# -> slope = -0.5, intercept = 1.5, y0 = 1   (SOLVED)
```

```
calculus_ai999x.solve('normal_line', f='exp(x)', a=0)
# -> slope = -1, intercept = 1, y0 = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the normal line to y = x^2 at x = 1. | SOLVED | slope = -0.5, intercept = 1.5, y0 = 1 |
| Find the equation of the normal line to f(x) = exp(x) at x = 0. | SOLVED | slope = -1, intercept = 1, y0 = 1 |

| question | outcome | result |
|---|---|---|
| Find the normal line to y = x^2 at x = 0. | INVALID_INPUT | Normal line: the tangent is horizontal there, so the normal line is vertical (x = a) and has no slope. |

### Average rate of change — `average_rate_of_change`

The average rate of change (f(b) - f(a))/(b - a) of f over [a, b]: the slope of the secant line.

$$\frac{\Delta f}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | start of the interval | |
| `b` | end of the interval | |

| output | meaning |
|---|---|
| `average_rate` | (f(b) - f(a))/(b - a) |
| `fa` | f(a) |
| `fb` | f(b) |

- **Method:** difference quotient of the two function values
- **Assumes:** $a \neq b$; f defined at a and b
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('average_rate_of_change', f='x^2', a=1, b=3)
# -> average_rate = 4, fa = 1, fb = 9   (SOLVED)
```

```
calculus_ai999x.solve('average_rate_of_change', f='sin(x)', a=0, b=1.5707963267948966)
# -> average_rate = 0.63662, fa = 0, fb = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the average rate of change of f(x) = x^2 from x = 1 to x = 3. | SOLVED | average_rate = 4, fa = 1, fb = 9 |
| What is the average rate of change of sin(x) on [0, pi/2]? | SOLVED | average_rate = 0.63662, fa = 0, fb = 1 |
| Find the slope of the secant line of x^3 between x = 0 and x = 2. | SOLVED | average_rate = 4, fa = 0, fb = 8 |
| Find the average rate of change of x^2 from x = 2 to x = 2. | INVALID_INPUT | Average rate of change: the two points must differ. |

### Velocity and acceleration from position — `velocity_acceleration`

Velocity $v = s'(t)$ and acceleration $a = s''(t)$ at time t0 from a position function s(t).

$$v = \frac{ds}{dt}, \quad a = \frac{d^2s}{dt^2}$$

| input | meaning | condition |
|---|---|---|
| `s` | the position s(t) | |
| `t0` | the time t0 | |

| output | meaning |
|---|---|
| `velocity` | v(t0) = s'(t0) |
| `acceleration` | a(t0) = s''(t0) |
| `speed` | \|v(t0)\| |
| `position` | s(t0) |

- **Method:** exact differentiation of the position once and twice
- **Assumes:** s twice differentiable at t0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('velocity_acceleration', s='t^3 - 6t^2 + 9t', t0=2)
# -> velocity = -3, acceleration = 0, speed = 3, position = 2   (SOLVED)
```

```
calculus_ai999x.solve('velocity_acceleration', s='5t^2 + 2t', t0=3)
# -> velocity = 32, acceleration = 10, speed = 32, position = 51   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A particle has position s(t) = t^3 - 6t^2 + 9t. Find its velocity and acceleration at t = 2. | SOLVED | velocity = -3, acceleration = 0, speed = 3, position = 2 |
| Find the velocity of s(t) = 5t^2 + 2t at t = 3. | SOLVED | velocity = 32, acceleration = 10, speed = 32, position = 51 |
| The position of an object is s(t) = 4sin(t). Find its velocity at t = 0. | SOLVED | velocity = 4, acceleration = 0, speed = 4, position = 0 |

## Linear approximation — `linear_approximation`

Estimate f(x1) by the tangent line at $x = a$: L(x1) = f(a) + f'(a)$(x_1 - a)$; with the actual value and error.

$$L(x) = f(a) + f'(a)\,(x - a)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the base point a | |
| `x1` | the point to estimate | |

| output | meaning |
|---|---|
| `approximation` | L(x1) |
| `actual` | f(x1) |
| `error` | f(x1) - L(x1) |
| `slope` | f'(a) |

- **Method:** linearization (tangent line) at the base point
- **Assumes:** f differentiable at a
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('linear_approximation', f='sqrt(x)', a=4, x1=4.1)
# -> approximation = 2.025, actual = 2.02485, error = -0.000154327, slope = 0.25   (SOLVED)
```

```
calculus_ai999x.solve('linear_approximation', f='exp(x)', a=0, x1=0.1)
# -> approximation = 1.1, actual = 1.10517, error = 0.00517092, slope = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Use the linear approximation of f(x) = sqrt(x) at x = 4 to estimate f(4.1). | SOLVED | approximation = 2.025, actual = 2.02485, error = -0.000154327, slope = 0.25 |
| Find the linearization of exp(x) at x = 0 and evaluate it at x = 0.1. | SOLVED | approximation = 1.1, actual = 1.10517, error = 0.00517092, slope = 1 |
| Use the tangent line approximation of x^3 at x = 2 to estimate f(2.1). | SOLVED | approximation = 9.2, actual = 9.261, error = 0.061, slope = 12 |

## Curvature of a graph — `curvature`

The curvature kappa = |f''| / (1 + f'^2)^(3/2) of $y = f(x)$ at $x = a$, and the radius of curvature $1/\kappa$.

$$\kappa = \frac{|f''(a)|}{(1 + f'(a)^2)^{3/2}}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `curvature` | kappa at $x = a$ |
| `radius_of_curvature` | $1/\kappa$ (absent where kappa = 0) |

- **Method:** curvature formula of a graph from the exact first and second derivatives
- **Assumes:** f twice differentiable at a
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('curvature', f='x^2', a=0)
# -> curvature = 2, radius_of_curvature = 0.5   (SOLVED)
```

```
calculus_ai999x.solve('curvature', f='sin(x)', a=1.5707963267948966)
# -> curvature = 1, radius_of_curvature = 1   (SOLVED)
```

```
calculus_ai999x.solve('curvature', f='x^2', a=1)
# -> curvature = 0.178885, radius_of_curvature = 5.59017   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the curvature of y = x^2 at x = 0. | SOLVED | curvature = 2, radius_of_curvature = 0.5 |
| What is the curvature of f(x) = sin(x) at x = pi/2? | SOLVED | curvature = 1, radius_of_curvature = 1 |
| Find the radius of curvature of y = x^2 at x = 0. | SOLVED | curvature = 2, radius_of_curvature = 0.5 |

## Definite integral — `definite_integral`

The definite integral of f(x) from $x = a$ to $x = b$ (net signed area).

$$\int_a^b f(x)\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(x) | |
| `a` | lower limit | |
| `b` | upper limit | |

| output | meaning |
|---|---|
| `integral` | the value of the integral |

- **Method:** high-precision tanh-sinh quadrature (30 digits) after an exact continuity check on [a, b]
- **Assumes:** f real and continuous on the closed interval between a and b
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('definite_integral', f='x^2', a=0, b=3)
# -> integral = 9   (SOLVED)
```

```
calculus_ai999x.solve('definite_integral', f='sin(x)', a=0, b=3.141592653589793)
# -> integral = 2   (SOLVED)
```

```
calculus_ai999x.solve('definite_integral', f='exp(2x)', a=0, b=1)
# -> integral = 3.19453   (SOLVED)
```

```
calculus_ai999x.solve('definite_integral', f='1/x', a=1, b=2.718281828459045)
# -> integral = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the integral of x^2 from 0 to 3. | SOLVED | integral = 9 |
| Find the definite integral of sin(x) from 0 to pi. | SOLVED | integral = 2 |
| Integrate exp(2x) from x = 0 to x = 1. | SOLVED | integral = 3.19453 |
| Compute the integral of 3x^2 + 2x dx from 1 to 2. | SOLVED | integral = 10 |
| Evaluate the integral of 1/x from -1 to 1. | INVALID_INPUT | Definite integral: the function is not continuous (or not real) on the whole interval [-1, 1]. |

## Average value of a function — `average_value`

The average value (1/(b - a)) * integral of f from a to b of a function on [a, b].

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | start of the interval | |
| `b` | end of the interval | |

| output | meaning |
|---|---|
| `average_value` | the average value of f on [a, b] |
| `integral` | the integral of f over [a, b] |

- **Method:** definite integral by high-precision quadrature, divided by the interval length
- **Assumes:** $a < b$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('average_value', f='x^2', a=0, b=3)
# -> average_value = 3, integral = 9   (SOLVED)
```

```
calculus_ai999x.solve('average_value', f='sin(x)', a=0, b=3.141592653589793)
# -> average_value = 0.63662, integral = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the average value of f(x) = x^2 on [0, 3]. | SOLVED | average_value = 3, integral = 9 |

| question | outcome | result |
|---|---|---|
| What is the average value of sin(x) from 0 to pi? | SOLVED | average_value = 0.63662, integral = 2 |
| Find the average value of x^2 on [3, 0]. | INVALID_INPUT | Average value of a function: the interval needs a < b. |

### Area between two curves — `area_between_curves`

The area between the graphs of two polynomials $y = f(x)$ and $y = g(x)$ over [a, b], counting every piece as positive (the curves may cross).

$$A = \int_a^b |f(x) - g(x)|\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the first curve f(x) (a polynomial) | |
| `g` | the second curve g(x) (a polynomial) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `area` | the area between the curves |
| `net_integral` | integral of f - g (signed) |
| `crossings` | x where the curves cross inside (a, b) |

- **Method:** exact integration of |f - g|, split at the exact real roots of f - g
- **Assumes:** $a < b$; f and g polynomials with rational coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('area_between_curves', f='x^2', g='x', a=0, b=2)
# -> area = 1, net_integral = 0.666667, crossings = [1.0]   (SOLVED)
```

```
calculus_ai999x.solve('area_between_curves', f='4 - x^2', g='x + 2', a=-2, b=1)
# -> area = 4.5, net_integral = 4.5, crossings = []   (SOLVED)
```

```
calculus_ai999x.solve('area_between_curves', f='x^3', g='0', a=-1, b=1)
# -> area = 0.5, net_integral = 0, crossings = [0.0]   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the area between y = x^2 and y = x from x = 0 to x = 2. | SOLVED | area = 1, net_integral = 0.666667, crossings = [1.0] |
| Find the area between f(x) = 4 - x^2 and g(x) = x + 2 on [-2, 1]. | SOLVED | area = 4.5, net_integral = 4.5, crossings = [] |
| Find the area between y = sin(x) and y = 0 from x = 0 to x = pi. | INVALID_INPUT | Area between two curves: this capability needs a polynomial. |

### Area enclosed by two curves — `area_enclosed_curves`

The area of the region enclosed by two polynomial curves $y = f(x)$ and $y = g(x)$, between their first and last intersection.

$$A = \int_{x_1}^{x_n} |f(x) - g(x)|\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the first curve f(x) (a polynomial) | |
| `g` | the second curve g(x) (a polynomial) | |

| output | meaning |
|---|---|
| `area` | the enclosed area |
| `x_left` | first intersection |
| `x_right` | last intersection |
| `intersections` | all intersection x-values |

- **Method:** exact real roots of f - g, then exact integration of |f - g| between consecutive roots
- **Assumes:** f and g polynomials with rational coefficients meeting in at least two points
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('area_enclosed_curves', f='x^2', g='x')
# -> area = 0.166667, x_left = 0, x_right = 1, intersections = [0.0, 1.0]   (SOLVED)
```

```
calculus_ai999x.solve('area_enclosed_curves', f='4 - x^2', g='0')
# -> area = 10.6667, x_left = -2, x_right = 2, intersections = [-2.0, 2.0]   (SOLVED)
```

```
calculus_ai999x.solve('area_enclosed_curves', f='x^3', g='x')
# -> area = 0.5, x_left = -1, x_right = 1, intersections = [-1.0, 0.0, 1.0]   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the area enclosed by y = x^2 and y = x. | SOLVED | area = 0.166667, x_left = 0, x_right = 1, intersections = [0.0, 1.0] |
| Find the area of the region bounded by y = 4 - x^2 and y = 0. | SOLVED | area = 10.6667, x_left = -2, x_right = 2, intersections = [-2.0, 2.0] |
| Find the area enclosed by the curves y = x^3 and y = x. | SOLVED | area = 0.5, x_left = -1, x_right = 1, intersections = [-1.0, 0.0, 1.0] |
| Find the area enclosed by y = x^2 + 1 and y = 0. | INVALID_INPUT | Area enclosed by two curves: the curves meet in fewer than two points, so they enclose no bounded region. |

### Volume of revolution (disk method) — `disk_volume`

The volume of the solid formed by rotating $y = f(x)$, $a \le x \le b$, about the x-axis.

$$V = \pi \int_a^b f(x)^2 \, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the curve f(x) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `volume` | the volume of the solid |

- **Method:** disk method: pi times the integral of $f^2$, by high-precision quadrature
- **Assumes:** $a < b$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('disk_volume', f='x^2', a=0, b=1)
# -> volume = 0.628319   (SOLVED)
```

```
calculus_ai999x.solve('disk_volume', f='sqrt(x)', a=0, b=4)
# -> volume = 25.1327   (SOLVED)
```

```
calculus_ai999x.solve('disk_volume', f='sqrt(4 - x^2)', a=-2, b=2)
# -> volume = 33.5103   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the volume when y = x^2 from x = 0 to x = 1 is rotated about the x-axis. | SOLVED | volume = 0.628319 |
| Find the volume of revolution of y = sqrt(x) about the x-axis on [0, 4]. | SOLVED | volume = 25.1327 |
| Find the volume of the solid obtained by rotating y = sqrt(4 - x^2) about the x-axis from x = -2 to x = 2. | SOLVED | volume = 33.5103 |

## Volume of revolution (washer method) — `washer_volume`

The volume swept when the region between an outer curve $y = f$(x) and an inner curve $y = g$(x) (polynomials, f >= g >= 0) on [a, b] is rotated about the x-axis.

$$V = \pi \int_a^b \left( f(x)^2 - g(x)^2 \right) dx$$

| input | meaning | condition |
|---|---|---|
| `f` | outer radius f(x) (a polynomial) | |
| `g` | inner radius g(x) (a polynomial) | |
| `a` | left end | |

| input | meaning | condition |
|---|---|---|
| `b` | right end | |

| output | meaning |
|---|---|
| `volume` | the volume of the solid |

- **Method:** washer method, integrated exactly
- **Assumes:** $a < b$; $f \geq g \geq 0$ on [a, b] (checked exactly); polynomials
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('washer_volume', f='x + 2', g='1', a=0, b=1)
# -> volume = 16.7552   (SOLVED)
```

```
calculus_ai999x.solve('washer_volume', f='4 - x^2', g='1', a=0, b=1)
# -> volume = 39.3746   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the volume when the region between y = x + 2 and y = 1 on [0, 1] is rotated about the x-axis using washers. | SOLVED | volume = 16.7552 |
| Use the washer method for outer radius 4 - x^2 and inner radius 1 on [0, 1]. | SOLVED | volume = 39.3746 |
| Use the washer method for outer radius 1 and inner radius x^2 + 2 on [0, 1]. | INVALID_INPUT | Volume of revolution (washer method): the washer method needs outer radius f >= inner radius g >= 0 on [a, b]. |

## Volume of revolution (shell method) — `shell_volume`

The volume swept when the region under $y = f$(x) >= 0 (a polynomial), $a \leq x \leq b$ with $a \geq 0$, is rotated about the y-axis.

$$V = 2\pi \int_a^b x\, f(x)\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the curve f(x) (a polynomial) | |
| `a` | left end (>= 0) | |
| `b` | right end | |

| output | meaning |
|---|---|
| `volume` | the volume of the solid |

- **Method:** cylindrical shells, integrated exactly
- **Assumes:** $0 \leq a < b$; f(x) >= 0 on [a, b] (checked exactly)

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('shell_volume', f='x^2', a=0, b=2)
# -> volume = 25.1327   (SOLVED)
```

```
calculus_ai999x.solve('shell_volume', f='4 - x^2', a=0, b=2)
# -> volume = 25.1327   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Use the shell method to find the volume when y = x^2 on [0, 2] is rotated about the y-axis. | SOLVED | volume = 25.1327 |
| Find the volume of the solid formed by rotating the region under y = 4 - x^2 from x = 0 to x = 2 about the y-axis. | SOLVED | volume = 25.1327 |
| Use the shell method to find the volume when y = x - 3 on [0, 2] is rotated about the y-axis. | INVALID_INPUT | Volume of revolution (shell method): the shell method needs f(x) >= 0 on [a, b]. |

## Arc length of a graph — `arc_length`

The length of the curve $y = f$(x) from $x = a$ to $x = b$.

$$L = \int_a^b \sqrt{1 + f'(x)^2}\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the curve f(x) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `arc_length` | the length of the curve |

- **Method:** arc-length integral of sqrt(1 + $f$'(x)^2) with the exact derivative, by high-precision quadrature
- **Assumes:** $a < b$; f' continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('arc_length', f='cosh(x)', a=0, b=1)
# -> arc_length = 1.1752   (SOLVED)
```

```
calculus_ai999x.solve('arc_length', f='2x + 1', a=0, b=3)
# -> arc_length = 6.7082   (SOLVED)
```

```
calculus_ai999x.solve('arc_length', f='x^2', a=0, b=1)
# -> arc_length = 1.47894   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the arc length of y = x^2 from x = 0 to x = 1. | SOLVED | arc_length = 1.47894 |
| What is the length of the curve y = cosh(x) on [0, 1]? | SOLVED | arc_length = 1.1752 |
| Find the arc length of y = sqrt(x) from x = -1 to x = 1. | INVALID_INPUT | Arc length of a graph: the function is not continuous (or not real) on the whole interval [-1, 1]. |

## Surface area of revolution — `surface_area_revolution`

The area of the surface formed by rotating $y = f(x)$, $a \le x \le b$, about the x-axis.

$$S = 2\pi \int_a^b |f(x)| \sqrt{1 + f'(x)^2}\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the curve f(x) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `surface_area` | the area of the surface of revolution |

- **Method:** $2pi$ times the integral of |f| * sqrt($1 + f'$^2) (split where f changes sign), high-precision quadrature
- **Assumes:** $a < b$; f' continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('surface_area_revolution', f='x', a=0, b=1)
# -> surface_area = 4.44288   (SOLVED)
```

```
calculus_ai999x.solve('surface_area_revolution', f='sqrt(4 - x^2)', a=-1, b=1)
# -> surface_area = 25.1327   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the surface area when y = x from x = 0 to x = 1 is rotated about the x-axis. | SOLVED | surface_area = 4.44288 |
| Find the area of the surface of revolution of y = sqrt(4 - x^2) about the x-axis on [-1, 1]. | SOLVED | surface_area = 25.1327 |

## Centroid of a region under a curve — `centroid_region`

The area and centroid (x_bar, y_bar) of the region between $y = f(x)$ >= 0 (a polynomial) and the x-axis for $a \le x \le b$.

$$\bar{x} = \frac{1}{A}\int_a^b x f(x)\,dx, \quad \bar{y} = \frac{1}{2A}\int_a^b f(x)^2\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the curve f(x) (a polynomial) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `area` | area of the region |
| `x_bar` | x-coordinate of the centroid |
| `y_bar` | y-coordinate of the centroid |

- **Method:** exact moments of the region: x_bar = (1/A) * integral of x f, y_bar = (1/2A) * integral of $f^2$
- **Assumes:** $a < b$; f(x) >= 0 on [a, b] (checked exactly)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('centroid_region', f='x^2', a=0, b=1)
# -> area = 0.333333, x_bar = 0.75, y_bar = 0.3   (SOLVED)
```

```
calculus_ai999x.solve('centroid_region', f='4 - x^2', a=-2, b=2)
# -> area = 10.6667, x_bar = 0, y_bar = 1.6   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the centroid of the region under y = x^2 from x = 0 to x = 1. | SOLVED | area = 0.333333, x_bar = 0.75, y_bar = 0.3 |
| Find the centroid of the region bounded by y = 4 - x^2 and the x-axis on [-2, 2]. | SOLVED | area = 10.6667, x_bar = 0, y_bar = 1.6 |

## Improper integral to infinity — `improper_integral`

The improper integral of f(x) from $x = a$ to infinity, when it converges.

$$\int_a^\infty f(x)\,dx = \lim_{R\to\infty}\int_a^R f(x)\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(x) | |
| `a` | lower limit | |

| output | meaning |
|---|---|
| `integral` | the value of the improper integral |

- **Method:** convergence by a limit comparison with $1/x^{1.01}$ (exact limit), value by tanh-sinh quadrature on [a, infinity)
- **Assumes:** f continuous on [a, infinity); $x^{1.01} f$(x) has a finite limit (comparison test), so the integral converges absolutely
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('improper_integral', f='exp(-x)', a=0)
# -> integral = 1   (SOLVED)
```

```
calculus_ai999x.solve('improper_integral', f='1/x^2', a=1)
# -> integral = 1   (SOLVED)
```

```
calculus_ai999x.solve('improper_integral', f='1/(x^2 + 1)', a=0)
# -> integral = 1.5708   (SOLVED)
```

```
calculus_ai999x.solve('improper_integral', f='exp(-x^2)', a=0)
# -> integral = 0.886227   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the improper integral of exp(-x) from 0 to infinity. | SOLVED | integral = 1 |
| Find the integral of 1/x^2 from 1 to infinity. | SOLVED | integral = 1 |
| Find the integral of 1/x from 1 to infinity. | INVALID_INPUT | Improper integral to infinity: convergence cannot be established (the integrand does not decay faster than 1/x^1.01). |
| Evaluate the integral of sin(x) from 0 to infinity. | INVALID_INPUT | Improper integral to infinity: convergence cannot be established (the integrand does not decay faster than 1/x^1.01). |

## Left Riemann sum — `left_riemann_sum`

The left-endpoint Riemann sum of f on [a, b] with n equal subintervals.

$$L_n = h \sum_{k=0}^{n-1} f(a + kh), \quad h = \frac{b-a}{n}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |
| `n` | number of equal subintervals | |

| output | meaning |
|---|---|
| `approximation` | the left Riemann sum |

- **Method:** h times the sum of f at the left endpoints $x_k = a + k$ h, $k = 0..n-1$ $(h = (b-a)/n)$
- **Assumes:** $a < b$; $1 \le n \le 100000$; f defined at the sample points
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('left_riemann_sum', f='x^2', a=0, b=2, n=4)
# -> approximation = 1.75   (SOLVED)
```

```
calculus_ai999x.solve('left_riemann_sum', f='x^3', a=0, b=1, n=4)
# -> approximation = 0.140625   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the left Riemann sum of x^2 on [0, 2] with n = 4. | SOLVED | approximation = 1.75 |
| Approximate the integral of x^3 from 0 to 1 using a left Riemann sum with 4 subintervals. | SOLVED | approximation = 0.140625 |
| Find the left Riemann sum of x^2 on [0, 2] with n = 0. | INVALID_INPUT | n: must be at least 1, got 0. |

## Right Riemann sum — `right_riemann_sum`

The right-endpoint Riemann sum of f on [a, b] with n equal subintervals.

$$R_n = h \sum_{k=1}^{n} f(a + kh), \quad h = \frac{b-a}{n}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |
| `n` | number of equal subintervals | |

| output | meaning |
|---|---|
| `approximation` | the right Riemann sum |

- **Method:** h times the sum of f at the right endpoints $x_k = a + k$ h, $k = 1..n$
- **Assumes:** $a < b$; $1 \le n \le 100000$; f defined at the sample points
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('right_riemann_sum', f='x^2', a=0, b=2, n=4)
# -> approximation = 3.75   (SOLVED)
```

```
calculus_ai999x.solve('right_riemann_sum', f='x^3', a=0, b=1, n=4)
# -> approximation = 0.390625   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the right Riemann sum of x^2 on [0, 2] with n = 4. | SOLVED | approximation = 3.75 |
| Approximate the integral of x^3 from 0 to 1 using a right Riemann sum with 4 subintervals. | SOLVED | approximation = 0.390625 |

## Trapezoidal rule — `trapezoid_rule`

The composite trapezoidal approximation of the integral of f on [a, b] with n subintervals, with the exact value and the error (approximation - exact).

$$T_n = \frac{h}{2}\left[f(x_0) + 2\sum_{k=1}^{n-1} f(x_k) + f(x_n)\right]$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |
| `n` | number of equal subintervals | |

| output | meaning |
|---|---|
| `approximation` | trapezoidal approximation T_n |
| `exact` | the integral |
| `error` | T_n - exact |

- **Method:** composite trapezoidal rule; exact value by high-precision quadrature
- **Assumes:** $a < b$; $1 \le n \le 100000$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('trapezoid_rule', f='x^2', a=0, b=2, n=4)
# -> approximation = 2.75, exact = 2.66667, error = 0.0833333   (SOLVED)
```

```
calculus_ai999x.solve('trapezoid_rule', f='1/x', a=1, b=2, n=4)
# -> approximation = 0.697024, exact = 0.693147, error = 0.00387663   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Approximate the integral of x^2 from 0 to 2 using the trapezoidal rule with n = 4. | SOLVED | approximation = 2.75, exact = 2.66667, error = 0.0833333 |
| Use the trapezoid rule with 4 subintervals to estimate the integral of 1/x from 1 to 2. | SOLVED | approximation = 0.697024, exact = 0.693147, error = 0.00387663 |
| Use the trapezoidal rule with 4 subintervals to estimate the integral of 1/x from -1 to 1. | INVALID_INPUT | Trapezoidal rule: the expression is not defined there (ZeroDivisionError). |

### Simpson's rule — `simpson_rule`

The composite Simpson approximation of the integral of f on [a, b] with an even number n of subintervals, with the exact value and the error.

$$S_n = \frac{h}{3}\left[f(x_0) + 4\sum_{\text{odd } k} f(x_k) + 2\sum_{\text{even } k} f(x_k) + f(x_n)\right]$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |
| `n` | number of equal subintervals | |

| output | meaning |
|---|---|
| `approximation` | Simpson approximation S_n |
| `exact` | the integral |
| `error` | S_n - exact |

- **Method:** composite Simpson's $1/3$ rule (weights 1, 4, 2, ..., 4, 1); exact value by high-precision quadrature
- **Assumes:** $a < b$; n even, $2 \le n \le 100000$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('simpson_rule', f='x^4', a=0, b=2, n=4)
# -> approximation = 6.41667, exact = 6.4, error = 0.0166667   (SOLVED)
```

```
calculus_ai999x.solve('simpson_rule', f='x^3', a=0, b=2, n=2)
# -> approximation = 4, exact = 4, error = 0   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Approximate the integral of x^4 from 0 to 2 using Simpson's rule with 4 subintervals. | SOLVED | approximation = 6.41667, exact = 6.4, error = 0.0166667 |
| Use Simpson's rule with n = 2 to approximate the integral of x^3 from 0 to 2. | SOLVED | approximation = 4, exact = 4, error = 0 |
| Use Simpson's rule with n = 3 to approximate the integral of x^3 from 0 to 2. | INVALID_INPUT | Simpson's rule: Simpson's rule needs an even number of subintervals. |

### Midpoint rule — `midpoint_rule`

The composite midpoint approximation of the integral of f on [a, b] with n subintervals, with the exact value and the error.

$$M_n = h \sum_{k=0}^{n-1} f\left(a + (k + \tfrac{1}{2})h\right)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |
| `n` | number of equal subintervals | |

| output | meaning |
|---|---|
| `approximation` | midpoint approximation M_n |
| `exact` | the integral |
| `error` | M_n - exact |

- **Method:** composite midpoint rule; exact value by high-precision quadrature
- **Assumes:** $a < b$; $1 \le n \le 100000$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('midpoint_rule', f='x^2', a=0, b=2, n=4)
# -> approximation = 2.625, exact = 2.66667, error = -0.0416667   (SOLVED)
```

```
calculus_ai999x.solve('midpoint_rule', f='x^3', a=0, b=2, n=2)
# -> approximation = 3.5, exact = 4, error = -0.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Approximate the integral of x^2 from 0 to 2 using the midpoint rule with n = 4. | SOLVED | approximation = 2.625, exact = 2.66667, error = -0.0416667 |
| Use the midpoint rule with 2 subintervals to estimate the integral of x^3 from 0 to 2. | SOLVED | approximation = 3.5, exact = 4, error = -0.5 |

## Gauss-Legendre quadrature — `gauss_legendre`

The n-point Gauss-Legendre approximation of the integral of f on [a, b] (exact for polynomials of degree <= 2n - 1), with the exact value and the error.

$$\int_a^b f\,dx \approx \frac{b-a}{2} \sum_{i=1}^{n} w_i\, f\left(\frac{a+b}{2} + \frac{b-a}{2} x_i\right)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | left end | |
| `b` | right end | |

| input | meaning | condition |
|---|---|---|
| `n` | number of nodes | |

| output | meaning |
|---|---|
| `approximation` | n-point Gauss-Legendre value |
| `exact` | the integral |
| `error` | approximation - exact |

- **Method:** nodes = roots of the Legendre polynomial P_n (Newton's method on the three-term recurrence), weights 2/((1 − x²) P_n'(x)^2), mapped to [a, b]
- **Assumes:** $a < b$; $1 \le n \le 64$; f continuous on [a, b]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('gauss_legendre', f='x^3 + x^2', a=0, b=2, n=2)
# -> approximation = 6.66667, exact = 6.66667, error = 0   (SOLVED)
```

```
calculus_ai999x.solve('gauss_legendre', f='x^4', a=-1, b=1, n=2)
# -> approximation = 0.222222, exact = 0.4, error = -0.177778   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Use 2-point Gauss-Legendre quadrature to approximate the integral of x^3 + x^2 from 0 to 2. | SOLVED | approximation = 6.66667, exact = 6.66667, error = 0 |
| Use 2-point Gaussian quadrature to approximate the integral of x^4 from -1 to 1. | SOLVED | approximation = 0.222222, exact = 0.4, error = -0.177778 |

## Finite-difference derivative — `numerical_derivative`

A forward, backward or central difference approximation of f'(a) with step h, with the exact derivative and the error.

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |
| `h` | the step size $h > 0$ | |
| `method` | forward, backward or central difference | |

| output | meaning |
|---|---|
| `approximation` | the difference quotient |
| `exact` | the exact f'(a) |
| `error` | approximation - exact |

- **Method:** difference quotients (f(a+h)-f(a))/h, (f(a)-f(a-h))/h, (f(a+h)-f(a-h))/(2h) at 30 digits; exact derivative symbolically
- **Assumes:** $h > 0$; f defined at the points used
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('numerical_derivative', f='x^2', a=1, h=0.1, method='central')
# -> approximation = 2, exact = 2, error = 0   (SOLVED)
```

```
calculus_ai999x.solve('numerical_derivative', f='x^3', a=1, h=0.5, method='forward')
# -> approximation = 4.75, exact = 3, error = 1.75   (SOLVED)
```

```
calculus_ai999x.solve('numerical_derivative', f='x^3', a=1, h=0.5, method='backward')
# -> approximation = 1.75, exact = 3, error = -1.25   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Estimate the derivative of x^2 at x = 1 using a central difference with h = 0.1. | SOLVED | approximation = 2, exact = 2, error = 0 |
| Use a forward difference with h = 0.5 to approximate the derivative of x^3 at x = 1. | SOLVED | approximation = 4.75, exact = 3, error = 1.75 |
| Use a backward difference with h = 0.5 to approximate the derivative of x^3 at x = 1. | SOLVED | approximation = 1.75, exact = 3, error = -1.25 |
| Use a central difference with h = -0.1 to approximate the derivative of x^3 at x = 1. | INVALID_INPUT | Finite-difference derivative: the step h must be positive. |

## Newton's method — `newton_method`

n iterations of Newton's method x_{$k+1$} = x_k - f(x_k)/f'(x_k) for a root of f, from x_0.

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `x0` | the starting value x_0 | |
| `n` | number of iterations | |

| output | meaning |
|---|---|
| `x_n` | the n-th iterate |
| `f_x_n` | f(x_n) |
| `iterates` | x_1 .. x_n |

- **Method:** Newton-Raphson with the exact derivative, at 30 digits
- **Assumes:** f'$(x_k) \neq 0$ at every iterate; $1 \leq n \leq 50$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('newton_method', f='x^2 - 2', x0=1, n=3)
# -> x_n = 1.41422, f_x_n = 6.0073e-06, iterates = [1.5, 1.4166666666666667, 1.4142156862745099]
(SOLVED)
```

```
calculus_ai999x.solve('newton_method', f='x^3 - 2x - 5', x0=2, n=1)
# -> x_n = 2.1, f_x_n = 0.061, iterates = [2.1]   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Apply Newton's method to f(x) = x^2 - 2 starting at x0 = 1 for 3 iterations. | SOLVED | x_n = 1.41422, f_x_n = 6.0073e-06, iterates = [1.5, 1.4166666666666667, 1.4142156862745099] |
| Use 1 iteration of Newton's method on x^3 - 2x - 5 from x0 = 2. | SOLVED | x_n = 2.1, f_x_n = 0.061, iterates = [2.1] |
| Apply Newton's method to f(x) = x^2 - 2 starting at x0 = 0 for 3 iterations. | INVALID_INPUT | Newton's method: f'(x) = 0 at an iterate, so Newton's method cannot continue. |

### Limit at a point — `limit_at_point`

The two-sided limit of f(x) as x approaches a (including removable $0/0$ forms such as sin(x)/x).

$$\lim_{x \to a} f(x)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the point $x = a$ | |

| output | meaning |
|---|---|
| `limit` | the limit of f(x) as x -> a |

- **Method:** exact symbolic limit (series expansion / Gruntz algorithm) from both sides; refused if the one-sided limits differ or are infinite
- **Assumes:** the left and right limits exist, are finite and agree
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('limit_at_point', f='(x^2 - 4)/(x - 2)', a=2)
# -> limit = 4   (SOLVED)
```

```
calculus_ai999x.solve('limit_at_point', f='sin(3x)/x', a=0)
# -> limit = 3   (SOLVED)
```

```
calculus_ai999x.solve('limit_at_point', f='(1 - cos(x))/x^2', a=0)
# -> limit = 0.5   (SOLVED)
```

```
calculus_ai999x.solve('limit_at_point', f='(exp(2x) - 1)/x', a=0)
# -> limit = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the limit of (x^2 - 4)/(x - 2) as x approaches 2. | SOLVED | limit = 4 |
| What is the limit of sin(3x)/x as x -> 0? | SOLVED | limit = 3 |
| Evaluate the limit as x approaches 0 of (1 - cos(x))/x^2. | SOLVED | limit = 0.5 |
| Find the limit of 1/x as x approaches 0. | INVALID_INPUT | Limit at a point: the limit is infinite (it does not exist as a finite number). |
| Find the limit of 1/x^2 as x approaches 0. | INVALID_INPUT | Limit at a point: the limit is infinite (it does not exist as a finite number). |

## Limit at infinity — `limit_at_infinity`

The limit of f(x) as x grows without bound (horizontal asymptote, end behaviour).

$$\lim_{x \to \infty} f(x)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |

| output | meaning |
|---|---|
| `limit` | the limit of f(x) as x -> infinity |

- **Method:** exact symbolic limit at +infinity (Gruntz algorithm); refused if infinite or oscillating
- **Assumes:** the limit exists and is finite
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('limit_at_infinity', f='(3x^2 + 1)/(x^2 - 5)')
# -> limit = 3   (SOLVED)
```

```
calculus_ai999x.solve('limit_at_infinity', f='(1 + 2/x)^x')
# -> limit = 7.38906   (SOLVED)
```

```
calculus_ai999x.solve('limit_at_infinity', f='sqrt(x^2 + 4x) - x')
# -> limit = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the limit of (3x^2 + 1)/(x^2 - 5) as x approaches infinity. | SOLVED | limit = 3 |

| question | outcome | result |
|---|---|---|
| What is the limit of (1 + 2/x)^x as x -> infinity? | SOLVED | limit = 7.38906 |
| Find the limit as x approaches infinity of sqrt(x^2 + 4x) - x. | SOLVED | limit = 2 |
| Find the limit of x^2 as x approaches infinity. | INVALID_INPUT | Limit at infinity: the limit is infinite (it does not exist as a finite number). |
| Find the limit of sin(x) as x approaches infinity. | INVALID_INPUT | Limit at infinity: the limit does not exist or could not be determined. |

### Implicit differentiation — `implicit_derivative`

The slope dy/dx of a curve given implicitly by F(x, y) = c, at a point on the curve.

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

| input | meaning | condition |
|---|---|---|
| `F` | the curve as an equation in x and y | |
| `p` | the point (x, y) on the curve | |

| output | meaning |
|---|---|
| `dy_dx` | dy/dx at the point |

- **Method:** $dy/dx = -F_x/F_y$ with exact symbolic partial derivatives of F(x, y) - c
- **Assumes:** the point lies on the curve; $F_y \neq 0$ there (no vertical tangent)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('implicit_derivative', F='x^2 + y^2 = 25', p='3, 4')
# -> dy_dx = -0.75   (SOLVED)
```

```
calculus_ai999x.solve('implicit_derivative', F='x^3 + y^3 = 9', p=[1, 2])
# -> dy_dx = -0.25   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find dy/dx for x^2 + y^2 = 25 at the point (3, 4). | SOLVED | dy_dx = -0.75 |
| Use implicit differentiation to find dy/dx for x^3 + y^3 = 9 at (1, 2). | SOLVED | dy_dx = -0.25 |
| Find dy/dx for x^2 + y^2 = 25 at the point (1, 1). | INVALID_INPUT | Implicit differentiation: the point (1, 1) is not on the curve. |
| Find dy/dx for x^2 + y^2 = 25 at the point (5, 0). | INVALID_INPUT | Implicit differentiation: F_y = 0 at this point: the tangent is vertical (dy/dx is undefined). |

### Absolute extrema on a closed interval — `extrema_on_interval`

The absolute maximum and minimum values of a polynomial on [a, b] by the closed-interval method, with the critical points inside (a, b).

$$\max_{[a,b]} f = \max\{f(a), f(b), f(c) : f'(c) = 0\}$$

| input | meaning | condition |
|---|---|---|
| `f` | the polynomial f(x) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `absolute_max` | the largest value of f on [a, b] |
| `absolute_min` | the smallest value |
| `critical_points` | critical points in (a, b), ascending |

- **Method:** exact real roots of f' in (a, b) (Sturm isolation), then f compared at the critical points and both ends
- **Assumes:** $a < b$; f is a polynomial with rational coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('extrema_on_interval', f='x^3 - 3x', a=0, b=2)
# -> absolute_max = 2, absolute_min = -2, critical_points = [1.0]   (SOLVED)
```

```
calculus_ai999x.solve('extrema_on_interval', f='x^2 - 4x + 1', a=-1, b=3)
# -> absolute_max = 6, absolute_min = -3, critical_points = [2.0]   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the absolute maximum and minimum of x^3 - 3x on [0, 2]. | SOLVED | absolute_max = 2, absolute_min = -2, critical_points = [1.0] |
| Find the absolute extrema of x^2 - 4x + 1 on the interval [-1, 3]. | SOLVED | absolute_max = 6, absolute_min = -3, critical_points = [2.0] |
| Find the absolute extrema of x^2 on the interval [3, 1]. | INVALID_INPUT | Absolute extrema on a closed interval: the interval needs a < b. |
| Find the absolute extrema of sin(x) on the interval [0, 3]. | INVALID_INPUT | Absolute extrema on a closed interval: this capability needs a polynomial. |

### Critical points and their classification — `critical_points_classify`

All critical points of a polynomial f(x) and which are local maxima or minima (first-derivative test).

$$f'(c) = 0,\ f' \text{ changes } + \to - \Rightarrow \text{local max}$$

| input | meaning | condition |
|---|---|---|
| `f` | the polynomial f(x) | |

| output | meaning |
|---|---|
| `critical_points` | real solutions of f'$(x) = 0$, ascending |
| `local_maxima` | critical points that are local maxima |
| `local_minima` | critical points that are local minima |

- **Method:** exact real roots of f' (Sturm isolation); the sign of f' on each side decides max / min / neither
- **Assumes:** f is a polynomial of degree >= 2 with rational coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('critical_points_classify', f='x^3 - 3x')
# -> critical_points = [-1.0, 1.0], local_maxima = [-1.0], local_minima = [1.0]   (SOLVED)
```

```
calculus_ai999x.solve('critical_points_classify', f='x^4 - 2x^2')
# -> critical_points = [-1.0, 0.0, 1.0], local_maxima = [0.0], local_minima = [-1.0, 1.0]
(SOLVED)
```

```
calculus_ai999x.solve('critical_points_classify', f='x^3')
# -> critical_points = [0.0], local_maxima = [], local_minima = []   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find and classify the critical points of x^3 - 3x. | SOLVED | critical_points = [-1.0, 1.0], local_maxima = [-1.0], local_minima = [1.0] |
| Find the local extrema of x^4 - 2x^2. | SOLVED | critical_points = [-1.0, 0.0, 1.0], local_maxima = [0.0], local_minima = [-1.0, 1.0] |
| Classify the critical points of 2x + 1. | INVALID_INPUT | Critical points and their classification: needs a polynomial of degree 2 or more (a line has no isolated critical points |

## Inflection points — `inflection_points`

The points where a polynomial changes concavity (f'' changes sign).

$$f''(c) = 0 \text{ and } f'' \text{ changes sign at } c$$

| input | meaning | condition |
|---|---|---|
| `f` | the polynomial f(x) | |

| output | meaning |
|---|---|
| `inflection_points` | x-values of the inflection points, ascending |

- **Method:** exact real roots of f''; kept only where f'' changes sign (odd multiplicity)
- **Assumes:** f is a polynomial of degree >= 2 with rational coefficients

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('inflection_points', f='x^3 - 3x^2')
# -> inflection_points = [1.0]   (SOLVED)
```

```
calculus_ai999x.solve('inflection_points', f='x^4 - 6x^2')
# -> inflection_points = [-1.0, 1.0]   (SOLVED)
```

```
calculus_ai999x.solve('inflection_points', f='x^4')
# -> inflection_points = []   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the inflection points of x^3 - 3x^2. | SOLVED | inflection_points = [1.0] |
| Find the points of inflection of x^4 - 6x^2. | SOLVED | inflection_points = [-1.0, 1.0] |
| Find the inflection points of x^4. | SOLVED | inflection_points = [] |

### Mean value theorem — `mean_value_theorem`

The secant slope of a polynomial on [a, b] and every c in (a, b) where f'(c) equals it.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

| input | meaning | condition |
|---|---|---|
| `f` | the polynomial f(x) | |
| `a` | left end | |
| `b` | right end | |

| output | meaning |
|---|---|
| `secant_slope` | (f(b) - f(a))/(b - a) |
| `c_values` | all c in (a, b) with f'(c) = the slope |

- **Method:** exact secant slope, then the exact real roots of f'(x) - slope in (a, b)
- **Assumes:** $a < b$; f a non-linear polynomial with rational coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('mean_value_theorem', f='x^2', a=0, b=2)
# -> secant_slope = 2, c_values = [1.0]   (SOLVED)
```

```
calculus_ai999x.solve('mean_value_theorem', f='x^3', a=0, b=3)
# -> secant_slope = 9, c_values = [1.7320508075688772]   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the value of c guaranteed by the mean value theorem for x^3 on [0, 3]. | SOLVED | secant_slope = 9, c_values = [1.7320508075688772] |
| Apply the mean value theorem to x^2 on [0, 2]. | SOLVED | secant_slope = 2, c_values = [1.0] |
| Apply the mean value theorem to 3x + 1 on [0, 2]. | INVALID_INPUT | Mean value theorem: f is linear: every c in (a, b) satisfies the mean value theorem. |

### Derivative of an integral (fundamental theorem) — `ftc_derivative`

d/dx of the integral of f(t) from a to g(x), evaluated at $x = x_0$ (first fundamental theorem with the chain rule).

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))\,g'(x)$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(t) | |
| `a` | the constant lower limit | |
| `g` | the upper limit g(x) | |
| `x0` | the point $x = x_0$ | |

| output | meaning |
|---|---|
| `derivative` | d/dx of the integral at x0 |

- **Method:** f(g(x0)) * g'(x0) with the exact derivative of g
- **Assumes:** f continuous between a and g(x0); g differentiable at x0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('ftc_derivative', f='sin(t)', a=0, g='x^2', x0=1)
# -> derivative = 1.68294   (SOLVED)
```

```
calculus_ai999x.solve('ftc_derivative', f='t^2 + 1', a=1, g='3x', x0=2)
# -> derivative = 111   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the derivative of the integral from 0 to x^2 of sin(t) dt at x = 1. | SOLVED | derivative = 1.68294 |
| What is d/dx of the integral from 1 to 3x of t^2 + 1 dt at x = 2? | SOLVED | derivative = 111 |
| Find the derivative of the integral from 1 to x of 1/t dt at x = -1. | INVALID_INPUT | Derivative of an integral (fundamental theorem): the function is not continuous (or not real) on the whole interval [-1, |

### Sum of an infinite geometric series — `geometric_series_sum`

The sum a + a r + a $r^2 + \ldots$ of a convergent geometric series.

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

| input | meaning | condition |
|---|---|---|
| `a` | the first term | |
| `r` | the common ratio | |

| output | meaning |
|---|---|
| `sum` | the sum of the series |

- **Method:** $a/(1-r)$, exact in the stated decimals
- **Assumes:** |r| < 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('geometric_series_sum', a=3, r=0.5)
# -> sum = 6   (SOLVED)
```

```
calculus_ai999x.solve('geometric_series_sum', a=1, r=-0.25)
# -> sum = 0.8   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the sum of the geometric series with first term 3 and ratio 0.5. | SOLVED | sum = 6 |
| What is the sum of the infinite geometric series with first term 1 and common ratio -1/3? | SOLVED | sum = 0.75 |
| Find the sum of the geometric series with first term 3 and ratio 2. | INVALID_INPUT | Sum of an infinite geometric series: an infinite geometric series converges only when \|r\| < 1. |

## Partial sum of a geometric series — `geometric_partial_sum`

The sum of the first n terms a + a r + ... + a $r^{n-1}$.

$$S_n = a\,\frac{1-r^n}{1-r}$$

| input | meaning | condition |
|---|---|---|
| `a` | the first term | |
| `r` | the common ratio | |
| `n` | the number of terms | |

| output | meaning |
|---|---|
| `sum` | the sum of the first n terms |

- **Method:** a $(1-r^n)/(1-r)$ (n a when r = 1), exact in the stated decimals

- **Assumes:** $1 \leq n \leq 500$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('geometric_partial_sum', a=2, r=3, n=4)
# -> sum = 80   (SOLVED)
```

```
calculus_ai999x.solve('geometric_partial_sum', a=1, r=0.5, n=3)
# -> sum = 1.75   (SOLVED)
```

```
calculus_ai999x.solve('geometric_partial_sum', a=5, r=1, n=6)
# -> sum = 30   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the sum of the first 4 terms of the geometric series with first term 2 and ratio 3. | SOLVED | sum = 80 |
| What is the sum of the first 3 terms of a geometric sequence with first term 1 and common ratio 0.5? | SOLVED | sum = 1.75 |

### Sum of an arithmetic series — `arithmetic_series_sum`

The sum of the first n terms of an arithmetic sequence, and its last term.

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

| input | meaning | condition |
|---|---|---|
| `a1` | the first term | |
| `d` | the common difference | |
| `n` | the number of terms | |

| output | meaning |
|---|---|
| `sum` | the sum of the first n terms |
| `last_term` | the n-th term |

- **Method:** $S_n = n$ (2 $a_1 + (n-1)$ d)/2, exact in the stated decimals
- **Assumes:** $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('arithmetic_series_sum', a1=2, d=3, n=10)
# -> sum = 155, last_term = 29   (SOLVED)
```

```
calculus_ai999x.solve('arithmetic_series_sum', a1=1, d=1, n=100)
# -> sum = 5050, last_term = 100   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the sum of the first 10 terms of the arithmetic series with first term 2 and common difference 3. | SOLVED | sum = 155, last_term = 29 |
| What is the sum of the first 100 terms of an arithmetic sequence with first term 1 and difference 1? | SOLVED | sum = 5050, last_term = 100 |

### Sum of an infinite series — `series_sum`

The exact sum of a convergent series sum_{$n = k$}^infinity a_n (telescoping, geometric-type, zeta values, alternating harmonic, ...).

$$\sum_{n=k}^{\infty} a_n$$

| input | meaning | condition |
|---|---|---|
| `a` | the general term a_n, in n | |
| `k` | the starting index | |

| output | meaning |
|---|---|
| `sum` | the sum of the series |

- **Method:** exact symbolic summation; refused if the series diverges or no closed form is found
- **Assumes:** the series converges; a closed form exists
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('series_sum', a='1/n^2', k=1)
# -> sum = 1.64493   (SOLVED)
```

```
calculus_ai999x.solve('series_sum', a='1/(n*(n + 1))', k=1)
# -> sum = 1   (SOLVED)
```

```
calculus_ai999x.solve('series_sum', a='(1/2)^n', k=0)
# -> sum = 2   (SOLVED)
```

```
calculus_ai999x.solve('series_sum', a='(-1)^(n + 1)/n', k=1)
# -> sum = 0.693147   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the sum of the series 1/n^2 from n = 1 to infinity. | SOLVED | sum = 1.64493 |
| Find the sum of 1/(n*(n + 1)) from n = 1 to infinity. | SOLVED | sum = 1 |
| Find the sum of the series 1/n from n = 1 to infinity. | INVALID_INPUT | Sum of an infinite series: the series diverges. |

### Ratio test — `ratio_test`

The limit L of |a_(n+1)/a_n| and the ratio test's verdict for the series sum a_n (converges if $L < 1$, diverges if L > 1).

$$L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

| input | meaning | condition |
|---|---|---|
| `a` | the general term a_n, in n | |

| output | meaning |
|---|---|
| `ratio_limit` | L = lim \|a_(n+1)/a_n\| |
| `verdict` | converges or diverges |

- **Method:** exact simplification of a_(n+1)/a_n and its symbolic limit as n -> infinity
- **Assumes:** L exists, is finite and is not 1 (L = 1 is inconclusive and refused)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('ratio_test', a='2^n/n^3')
# -> ratio_limit = 2, verdict = diverges   (SOLVED)
```

```
calculus_ai999x.solve('ratio_test', a='n^2/3^n')
# -> ratio_limit = 0.333333, verdict = converges   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Apply the ratio test to the series 2^n/n^3. | SOLVED | ratio_limit = 2, verdict = diverges |
| Use the ratio test on n^2/3^n. | SOLVED | ratio_limit = 0.333333, verdict = converges |
| Apply the ratio test to the series 1/n^2. | INVALID_INPUT | Ratio test: the ratio limit is 1: the ratio test is inconclusive. |

### Taylor polynomial — `taylor_polynomial`

The degree-n Taylor polynomial of f about $x = a$ (Maclaurin when a = 0): its coefficients, its value at x1, f(x1) and the truncation error.

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `a` | the centre $x = a$ | |
| `n` | the degree | |
| `x1` | the evaluation point | |

| output | meaning |
|---|---|
| `coefficients` | c_0 .. c_n with $c_k = f^k$(a)/k! |
| `value` | P_n(x1) |
| `function_value` | f(x1) |
| `error` | f(x1) - P_n(x1) |

- **Method:** exact symbolic derivatives at a divided by k!, then Horner evaluation in $(x_1 - a)$
- **Assumes:** f is n times differentiable at a and defined at x1; $0 \le n \le 12$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('taylor_polynomial', f='exp(x)', a=0, n=3, x1=0.5)
# -> coefficients = [1.0, 1.0, 0.5, 0.16666666666666666], value = 1.64583, function_value =
1.64872, error = 0.00288794   (SOLVED)
```

```
calculus_ai999x.solve('taylor_polynomial', f='cos(x)', a=0, n=4, x1=0.3)
# -> coefficients = [1.0, 0.0, -0.5, 0.0, 0.041666666666666664], value = 0.955337, function_value =
0.955336, error = -1.01087e-06   (SOLVED)
```

```
calculus_ai999x.solve('taylor_polynomial', f='log(x)', a=1, n=2, x1=1.5)
# -> coefficients = [0.0, 1.0, -0.5], value = 0.375, function_value = 0.405465, error = 0.0304651
(SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the Taylor polynomial of degree 3 of exp(x) about x = 0 and evaluate it at x = 0.5. | SOLVED | coefficients = [1.0, 1.0, 0.5, 0.16666666666666666], value = 1.64583, function_value = 1.64872, error = 0.00288794 |
| Evaluate the degree 4 Maclaurin polynomial of cos(x) at x = 0.3. | SOLVED | coefficients = [1.0, 0.0, -0.5, 0.0, 0.041666666666666664], value = 0.955337, function_value = 0.955336, error = -1.01087e-06 |
| Find the Taylor polynomial of degree 2 of log(x) about x = 0 and evaluate it at x = 0.5. | INVALID_INPUT | Taylor polynomial: the result is not finite. |

## Partial derivative — `partial_derivative`

The partial derivative of f(x, y) or f(x, y, z) with respect to one variable, at a point.

$$\frac{\partial f}{\partial x}(p)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) or f(x, y, z) | |
| `var` | the variable | |
| `p` | the point (x, y) or (x, y, z) | |

| output | meaning |
|---|---|
| `partial` | the partial derivative at the point |

- **Method:** exact symbolic partial differentiation, then evaluation at the point
- **Assumes:** f differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('partial_derivative', f='x^2*y + y^3', var='x', p='1, 2')
# -> partial = 4   (SOLVED)
```

```
calculus_ai999x.solve('partial_derivative', f='x^2*y + y^3', var='y', p=[1, 2])
# -> partial = 13   (SOLVED)
```

```
calculus_ai999x.solve('partial_derivative', f='x*y*z', var='z', p='1, 2, 3')
# -> partial = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the partial derivative of x^2*y + y^3 with respect to x at (1, 2). | SOLVED | partial = 4 |
| What is the partial derivative of x*y*z with respect to z at the point (1, 2, 3)? | SOLVED | partial = 2 |
| Find the partial derivative of x^2*y with respect to z at (1, 2). | INVALID_INPUT | Partial derivative: the variable must be one of x, y. |

## Gradient — `gradient`

The gradient vector of f(x, y) or f(x, y, z) at a point and its length (the greatest rate of increase).

$$\nabla f = (f_x, f_y, f_z)$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) or f(x, y, z) | |
| `p` | the point (x, y) or (x, y, z) | |

| output | meaning |
|---|---|
| `gradient` | the gradient vector |
| `magnitude` | its length |

- **Method:** exact symbolic partial derivatives at the point
- **Assumes:** f differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('gradient', f='x^2 + y^2', p='1, 2')
# -> gradient = [2.0, 4.0], magnitude = 4.47214   (SOLVED)
```

```
calculus_ai999x.solve('gradient', f='x*y*z', p=[1, 2, 3])
# -> gradient = [6.0, 3.0, 2.0], magnitude = 7   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the gradient of x^2 + y^2 at (1, 2). | SOLVED | gradient = [2.0, 4.0], magnitude = 4.47214 |
| What is the gradient of x*y*z at the point (1, 2, 3)? | SOLVED | gradient = [6.0, 3.0, 2.0], magnitude = 7 |
| Find the gradient of log(x) + y at (-1, 2). | INVALID_INPUT | Gradient: the expression is not real there. |

## Directional derivative — `directional_derivative`

The rate of change of f at a point in the direction of a vector u (u is normalised to unit length).

$$D_{\mathbf{u}} f = \nabla f \cdot \frac{\mathbf{u}}{|\mathbf{u}|}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) or f(x, y, z) | |
| `p` | the point | |
| `u` | the direction vector | |

| output | meaning |
|---|---|
| `directional_derivative` | D_u f at the point |

- **Method:** gradient (exact partial derivatives) dotted with u/|u|
- **Assumes:** $u \neq 0$ with as many components as p; f differentiable at p
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('directional_derivative', f='x^2*y', p='1, 2', u='3, 4')
# -> directional_derivative = 3.2   (SOLVED)
```

```
calculus_ai999x.solve('directional_derivative', f='x*y*z', p=[1, 2, 3], u=[1, 2, 2])
# -> directional_derivative = 5.33333   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the directional derivative of x^2*y at (1, 2) in the direction (3, 4). | SOLVED | directional_derivative = 3.2 |
| Find the directional derivative of x*y*z at the point (1, 2, 3) in the direction of the vector (1, 2, 2). | SOLVED | directional_derivative = 5.33333 |

| question | outcome | result |
|---|---|---|
| Find the directional derivative of x^2*y at (1, 2) in the direction (0, 0). | INVALID_INPUT | Directional derivative: the direction vector must not be zero. |

### Tangent plane — `tangent_plane`

The tangent plane $z = f_x$ x + f_y y + c to the surface $z = f$(x, y) at (x0, y0).

$$z = f(x_0, y_0) + f_x(x - x_0) + f_y(y - y_0)$$

| input | meaning | condition |
|---|---|---|
| `f` | the surface $z = f$(x, y) | |
| `p` | the point (x0, y0) | |

| output | meaning |
|---|---|
| `z0` | f(x0, y0) |
| `fx` | f_x(x0, y0) |
| `fy` | f_y(x0, y0) |
| `c` | the constant term c |

- **Method:** exact partial derivatives; $z = z_0 + f_x\,(x - x_0) + f_y\,(y - y_0)$ expanded
- **Assumes:** f differentiable at (x0, y0)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('tangent_plane', f='x^2 + y^2', p='1, 2')
# -> z0 = 5, fx = 2, fy = 4, c = -5   (SOLVED)
```

```
calculus_ai999x.solve('tangent_plane', f='x*y', p=[2, 3])
# -> z0 = 6, fx = 3, fy = 2, c = -6   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the tangent plane to z = x^2 + y^2 at (1, 2). | SOLVED | z0 = 5, fx = 2, fy = 4, c = -5 |
| Find the equation of the tangent plane to the surface z = x*y at the point (2, 3). | SOLVED | z0 = 6, fx = 3, fy = 2, c = -6 |

### Second derivative test (two variables) — `hessian_classification`

Classifies a critical point of f(x, y) as a local minimum, local maximum or saddle point from the Hessian.

$$D = f_{xx} f_{yy} - f_{xy}^2$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) | |
| `p` | the critical point (x, y) | |

| output | meaning |
|---|---|
| `fxx` | f_xx |
| `fxy` | f_xy |
| `fyy` | f_yy |
| `D` | $D = f_xx\ f_yy - f_xy^2$ |
| `classification` | local minimum, local maximum or saddle point |

- **Method:** exact second partial derivatives at the point; $D > 0$ and $f_xx > 0$ minimum, $D > 0$ and $f_xx < 0$ maximum, $D < 0$ saddle
- **Assumes:** the gradient is zero at the point; $D \neq 0$ (D = 0 is inconclusive and refused)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('hessian_classification', f='x^2 + 3y^2 - 2x', p='1, 0')
# -> fxx = 2, fxy = 0, fyy = 6, D = 12, classification = local minimum   (SOLVED)
```

```
calculus_ai999x.solve('hessian_classification', f='x^2 - y^2', p=[0, 0])
# -> fxx = 2, fxy = 0, fyy = -2, D = -4, classification = saddle point   (SOLVED)
```

```
calculus_ai999x.solve('hessian_classification', f='4 - x^2 - y^2 + x*y', p=[0, 0])
# -> fxx = -2, fxy = 1, fyy = -2, D = 3, classification = local maximum   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Classify the critical point of x^2 + 3y^2 - 2x at (1, 0). | SOLVED | fxx = 2, fxy = 0, fyy = 6, D = 12, classification = local minimum |
| Use the second derivative test for x^2 - y^2 at the point (0, 0). | SOLVED | fxx = 2, fxy = 0, fyy = -2, D = -4, classification = saddle point |
| Classify the critical point of x^2 + y^2 at (1, 0). | INVALID_INPUT | Second derivative test (two variables): (1, 0) is not a critical point (the gradient is (2, 0)). |
| Classify the critical point of x^4 + y^4 at (0, 0). | INVALID_INPUT | Second derivative test (two variables): D = 0: the second derivative test is inconclusive. |

## Laplacian — `laplacian`

The Laplacian f_xx + f_yy (+ f_zz) of a function at a point.

$$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) or f(x, y, z) | |
| `p` | the point | |

| output | meaning |
|---|---|
| `laplacian` | the Laplacian at the point |

- **Method:** exact second partial derivatives summed
- **Assumes:** f twice differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('laplacian', f='x^2 + y^2', p='1, 1')
# -> laplacian = 4   (SOLVED)
```

```
calculus_ai999x.solve('laplacian', f='x^2*y + z^3', p=[1, 2, 3])
# -> laplacian = 22   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the Laplacian of x^2*y + z^3 at (1, 2, 3). | SOLVED | laplacian = 22 |
| Find the Laplacian of x^2 + y^2 at the point (1, 1). | SOLVED | laplacian = 4 |

## Divergence — `divergence`

The divergence P_x + Q_y + R_z of a vector field F = (P, Q, R) at a point.

$$\nabla \cdot \mathbf{F} = P_x + Q_y + R_z$$

| input | meaning | condition |
|---|---|---|
| `P` | the x-component P(x, y, z) | |
| `Q` | the y-component Q(x, y, z) | |
| `R` | the z-component R(x, y, z) | |
| `p` | the point (x, y, z) | |

| output | meaning |
|---|---|
| `divergence` | div F at the point |

- **Method:** exact symbolic partial derivatives
- **Assumes:** the components are differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('divergence', P='x*y', Q='y*z', R='x*z', p='1, 2, 3')
# -> divergence = 6   (SOLVED)
```

```
calculus_ai999x.solve('divergence', P='x^2', Q='y^2', R='z^2', p=[1, 1, 1])
# -> divergence = 6   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the divergence of F = (x*y, y*z, x*z) at (1, 2, 3). | SOLVED | divergence = 6 |

| question | outcome | result |
|---|---|---|
| Find the divergence of the vector field F = (x^2, y^2, z^2) at the point (1, 1, 1). | SOLVED | divergence = 6 |

### Curl — `curl`

The curl of a vector field F = (P, Q, R) at a point.

$$\nabla \times \mathbf{F} = (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y)$$

| input | meaning | condition |
|---|---|---|
| `P` | the x-component P(x, y, z) | |
| `Q` | the y-component Q(x, y, z) | |
| `R` | the z-component R(x, y, z) | |
| `p` | the point (x, y, z) | |

| output | meaning |
|---|---|
| `curl` | curl F at the point |

- **Method:** exact symbolic partial derivatives: (R_y - Q_z, P_z - R_x, Q_x - P_y)
- **Assumes:** the components are differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('curl', P='x*y', Q='y*z', R='z*x', p='1, 2, 3')
# -> curl = [-2.0, -3.0, -1.0]   (SOLVED)
```

```
calculus_ai999x.solve('curl', P='-y', Q='x', R='z', p=[1, 2, 3])
# -> curl = [0.0, 0.0, 2.0]   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the curl of F = (x*y, y*z, z*x) at (1, 2, 3). | SOLVED | curl = [-2.0, -3.0, -1.0] |
| Find the curl of the vector field F = (-y, x, z) at the point (1, 2, 3). | SOLVED | curl = [0.0, 0.0, 2.0] |

### Jacobian determinant — `jacobian_determinant`

The Jacobian determinant d(x, y)/d(u, v) of a transformation $x = x$(u, v), $y = y$(u, v) at a point.

$$\frac{\partial(x, y)}{\partial(u, v)} = x_u y_v - x_v y_u$$

| input | meaning | condition |
|---|---|---|
| `xu` | x as a function of u and v | |
| `yv` | y as a function of u and v | |
| `p` | the point (u, v) | |

| output | meaning |
|---|---|
| `jacobian` | the Jacobian determinant at the point |

- **Method:** exact partial derivatives; x_u y_v - x_v y_u
- **Assumes:** x and y differentiable at the point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('jacobian_determinant', xu='u*cos(v)', yv='u*sin(v)', p='2, 0.5')
# -> jacobian = 2   (SOLVED)
```

```
calculus_ai999x.solve('jacobian_determinant', xu='u + v', yv='u - v', p=[1, 1])
# -> jacobian = -2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the Jacobian of x = u*cos(v), y = u*sin(v) at (2, 0.5). | SOLVED | jacobian = 2 |
| Find the Jacobian determinant of the transformation x = u + v, y = u - v at the point (1, 1). | SOLVED | jacobian = -2 |

## Double integral over a rectangle — `double_integral_rectangle`

The double integral of f(x, y) over the rectangle [a, b] x [c, d].

$$\int_a^b \int_c^d f(x, y)\, dy\, dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(x, y) | |
| `a` | x from | |
| `b` | x to | |
| `c` | y from | |
| `d` | y to | |

| output | meaning |
|---|---|
| `integral` | the value of the double integral |

- **Method:** product Gauss-Legendre rules of increasing order (24, 40, 64 nodes per direction) until two agree to $1e-12$
- **Assumes:** $a < b$, $c < d$; f continuous on the rectangle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('double_integral_rectangle', f='x*y', a=0, b=2, c=0, d=3)
# -> integral = 9   (SOLVED)
```

```
calculus_ai999x.solve('double_integral_rectangle', f='x^2 + y', a=0, b=1, c=0, d=2)
# -> integral = 2.66667   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the double integral of x*y over [0, 2] x [0, 3]. | SOLVED | integral = 9 |
| Evaluate the double integral of x^2 + y over the rectangle [0, 1] x [0, 2]. | SOLVED | integral = 2.66667 |
| Find the double integral of x*y over [2, 0] x [0, 3]. | INVALID_INPUT | Double integral over a rectangle: the rectangle needs a < b and c < d. |

## Double integral over a region — `double_integral_region`

The double integral of f(x, y) over the region $a \le x \le b$, g1(x) <= y <= g2(x).

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(x, y) | |
| `a` | x from | |
| `b` | x to | |
| `g1` | the lower curve $y = g_1$(x) | |
| `g2` | the upper curve $y = g_2$(x) | |

| output | meaning |
|---|---|
| `integral` | the value of the double integral |

- **Method:** iterated Gauss-Legendre rules of increasing order (inner rule mapped to [g1(x), g2(x)]) until two agree
- **Assumes:** $a < b$; $g_1 \le g_2$ on [a, b]; f, g1, g2 continuous
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('double_integral_region', f='x*y', a=0, b=1, g1='x^2', g2='x')
# -> integral = 0.0416667   (SOLVED)
```

```
calculus_ai999x.solve('double_integral_region', f='1', a=0, b=2, g1='0', g2='x^2')
# -> integral = 2.66667   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the double integral of x*y over the region 0 <= x <= 1, x^2 <= y <= x. | SOLVED | integral = 0.0416667 |

| question | outcome | result |
|---|---|---|
| Evaluate the double integral of 1 over 0 <= x <= 2, 0 <= y <= x^2. | SOLVED | integral = 2.66667 |
| Find the double integral of x*y over the region 0 <= x <= 1, x <= y <= x^2 - 1. | INVALID_INPUT | Double integral over a region: the lower boundary must lie below the upper boundary on [a, b]. |

### Triple integral over a box — `triple_integral_box`

The triple integral of f(x, y, z) over the box [x0, x1] x [y0, y1] x [z0, z1].

$$\iiint_B f(x, y, z)\, dV$$

| input | meaning | condition |
|---|---|---|
| `f` | the integrand f(x, y, z) | |
| `x0` | x from | |
| `x1` | x to | |
| `y0` | y from | |
| `y1` | y to | |
| `z0` | z from | |
| `z1` | z to | |

| output | meaning |
|---|---|
| `integral` | the value of the triple integral |

- **Method:** exact iterated integration for a polynomial integrand; otherwise product Gauss-Legendre rules of increasing order until two agree
- **Assumes:** $x_0 < x_1$, $y_0 < y_1$, $z_0 < z_1$; f continuous on the box
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('triple_integral_box', f='x*y*z', x0=0, x1=1, y0=0, y1=2, z0=0, z1=3)
# -> integral = 4.5   (SOLVED)
```

```
calculus_ai999x.solve('triple_integral_box', f='x^2 + y^2 + z^2', x0=0, x1=1, y0=0, y1=1, z0=0,
z1=1)
# -> integral = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the triple integral of x*y*z over [0, 1] x [0, 2] x [0, 3]. | SOLVED | integral = 4.5 |
| Evaluate the triple integral of x^2 + y^2 + z^2 over the box [0, 1] x [0, 1] x [0, 1]. | SOLVED | integral = 1 |

### Surface area of a graph — `surface_area_graph`

The area of the surface $z = f$(x, y) over the rectangle [a, b] x [c, d].

$$A = \iint_R \sqrt{1 + f_x^2 + f_y^2}\, dA$$

| input | meaning | condition |
|---|---|---|
| `f` | the surface $z = f$(x, y) | |
| `a` | x from | |
| `b` | x to | |
| `c` | y from | |
| `d` | y to | |

| output | meaning |
|---|---|
| `surface_area` | the surface area |

- **Method:** exact partial derivatives; Gauss-Legendre product rules of sqrt(1 + f_x^2 + f_y^2) until two orders agree
- **Assumes:** $a < b$, $c < d$; f continuously differentiable on the rectangle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('surface_area_graph', f='x + y', a=0, b=1, c=0, d=1)
# -> surface_area = 1.73205   (SOLVED)
```

```
calculus_ai999x.solve('surface_area_graph', f='3', a=0, b=2, c=0, d=2)
# -> surface_area = 4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the surface area of z = x + y over [0, 1] x [0, 1]. | SOLVED | surface_area = 1.73205 |
| Find the area of the surface z = 2x + 2y over the rectangle [0, 1] x [0, 2]. | SOLVED | surface_area = 6 |

## Line integral of a vector field — `line_integral_vector`

The line integral (work) of F = (P, Q) along the path $x = x$(t), $y = y$(t) from t0 to t1.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t_0}^{t_1} \big(P\,x'(t) + Q\,y'(t)\big)dt$$

| input | meaning | condition |
|---|---|---|
| `P` | the x-component P(x, y) | |
| `Q` | the y-component Q(x, y) | |
| `xt` | the path x(t) | |
| `yt` | the path y(t) | |
| `t0` | start parameter | |
| `t1` | end parameter | |

| output | meaning |
|---|---|
| `line_integral` | the integral of F . dr along the path |

- **Method:** F(r(t)) . r'(t) built exactly (symbolic r'), then tanh-sinh quadrature at 20 digits
- **Assumes:** the integrand is continuous for t between t0 and t1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('line_integral_vector', P='-y', Q='x', xt='cos(t)', yt='sin(t)', t0=0,
t1=6.283185307179586)
# -> line_integral = 6.28319   (SOLVED)
```

```
calculus_ai999x.solve('line_integral_vector', P='y', Q='x', xt='t', yt='t^2', t0=0, t1=1)
# -> line_integral = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the line integral of F = (-y, x) along x = cos(t), y = sin(t) from t = 0 to t = 2pi. | SOLVED | line_integral = 6.28319 |
| Compute the line integral of the vector field F = (y, x) along the curve x = t, y = t^2 from t = 0 to t = 1. | SOLVED | line_integral = 1 |

### Line integral with respect to arc length — `line_integral_scalar`

The integral of f(x, y) ds along the path $x = x(t)$, $y = y(t)$, $t_0 \le t \le t_1$ (mass of a wire, average along a curve).

$$\int_C f\,ds = \int_{t_0}^{t_1} f(x(t), y(t))\sqrt{x'(t)^2 + y'(t)^2}\,dt$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x, y) | |
| `xt` | the path x(t) | |
| `yt` | the path y(t) | |
| `t0` | start parameter | |
| `t1` | end parameter | |

| output | meaning |
|---|---|
| `line_integral` | the integral of f ds along the path |

- **Method:** f(r(t)) |r'(t)| with exact r'(t), then tanh-sinh quadrature
- **Assumes:** $t_0 < t_1$; the integrand continuous on [t0, t1]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('line_integral_scalar', f='x + y', xt='t', yt='t', t0=0, t1=1)
# -> line_integral = 1.41421   (SOLVED)
```

```
calculus_ai999x.solve('line_integral_scalar', f='1', xt='cos(t)', yt='sin(t)', t0=0,
t1=3.141592653589793)
# -> line_integral = 3.14159   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the line integral of x + y ds along x = t, y = t from t = 0 to t = 1. | SOLVED | line_integral = 1.41421 |
| Find the line integral of x + y ds along x = t, y = t from t = 1 to t = 0. | INVALID_INPUT | Line integral with respect to arc length: the parameter must run forward (t0 < t1) for an integral with respect to arc l |

### Green's theorem on a rectangle — `greens_theorem`

Both sides of Green's theorem for F = (P, Q) on the rectangle [a, b] x [c, d]: the counter-clockwise circulation around the boundary and the double integral of Q_x - P_y.

$$\oint_C P\,dx + Q\,dy = \iint_R (Q_x - P_y)dA$$

| input | meaning | condition |
|---|---|---|
| `P` | the x-component P(x, y) | |
| `Q` | the y-component Q(x, y) | |
| `a` | x from | |
| `b` | x to | |
| `c` | y from | |
| `d` | y to | |

| output | meaning |
|---|---|
| `circulation` | the boundary line integral (counter-clockwise) |
| `double_integral` | the integral of Q_x - P_y |

- **Method:** the four sides by tanh-sinh quadrature; the double integral of the exact Q_x - P_y by Gauss-Legendre product rules; refused unless the two agree
- **Assumes:** $a < b$, $c < d$; P and Q continuously differentiable on the rectangle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('greens_theorem', P='-y', Q='x', a=0, b=1, c=0, d=2)
# -> circulation = 4, double_integral = 4   (SOLVED)
```

```
calculus_ai999x.solve('greens_theorem', P='x*y', Q='x^2', a=0, b=1, c=0, d=1)
# -> circulation = 0.5, double_integral = 0.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Verify Green's theorem for F = (-y, x) on the rectangle [0, 1] x [0, 2]. | SOLVED | circulation = 4, double_integral = 4 |
| Use Green's theorem for F = (x*y, x^2) on [0, 1] x [0, 1]. | SOLVED | circulation = 0.5, double_integral = 0.5 |

### Outward flux through a box (divergence theorem) — `divergence_theorem_box`

The outward flux of F = (P, Q, R) through the surface of the box [x0, x1] x [y0, y1] x [z0, z1].

$$\oiint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_B \nabla \cdot \mathbf{F}\, dV$$

| input | meaning | condition |
|---|---|---|
| `P` | the x-component | |
| `Q` | the y-component | |
| `R` | the z-component | |
| `x0` | x from | |
| `x1` | x to | |
| `y0` | y from | |
| `y1` | y to | |
| `z0` | z from | |
| `z1` | z to | |

| output | meaning |
|---|---|
| `flux` | the outward flux |

- **Method:** the divergence theorem: the triple integral of the exact divergence (exact for polynomials, otherwise Gauss-Legendre product rules until two orders agree)
- **Assumes:** the box is non-degenerate; F continuously differentiable on the box
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('divergence_theorem_box', P='x^2', Q='y^2', R='z^2', x0=0, x1=1, y0=0, y1=1,
z0=0, z1=1)
# -> flux = 3   (SOLVED)
```

```
calculus_ai999x.solve('divergence_theorem_box', P='x', Q='y', R='z', x0=0, x1=1, y0=0, y1=2, z0=0,
z1=3)
# -> flux = 18   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the outward flux of F = (x^2, y^2, z^2) through the box [0, 1] x [0, 1] x [0, 1]. | SOLVED | flux = 3 |
| Find the flux of the field F = (x, y, z) through the box [0, 1] x [0, 2] x [0, 3]. | SOLVED | flux = 18 |

### Arc length of a parametric curve — `parametric_arc_length`

The length of the curve $x = x(t)$, $y = y(t)$ for $t_0 \le t \le t_1$.

$$L = \int_{t_0}^{t_1} \sqrt{x'(t)^2 + y'(t)^2}\, dt$$

| input | meaning | condition |
|---|---|---|
| `xt` | x(t) | |
| `yt` | y(t) | |
| `t0` | start parameter | |
| `t1` | end parameter | |

| output | meaning |
|---|---|
| `arc_length` | the length of the curve |

- **Method:** sqrt(x'$(t)^2 + y$'(t)^2) with exact derivatives, tanh-sinh quadrature
- **Assumes:** $t_0 < t_1$; x and y continuously differentiable
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('parametric_arc_length', xt='cos(t)', yt='sin(t)', t0=0,
t1=3.141592653589793)
# -> arc_length = 3.14159   (SOLVED)
```

```
calculus_ai999x.solve('parametric_arc_length', xt='3t', yt='4t', t0=0, t1=2)
# -> arc_length = 10   (SOLVED)
```

#### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the arc length of the parametric curve x = cos(t), y = sin(t) from t = 0 to t = pi. | SOLVED | arc_length = 3.14159 |
| Find the length of the curve x = 3t, y = 4t from t = 0 to t = 2. | SOLVED | arc_length = 10 |

### Slope and concavity of a parametric curve — `parametric_slope`

dy/dx and $d^{2y}/dx^2$ of the curve $x = x(t)$, $y = y(t)$ at a parameter value t0.

$$\frac{dy}{dx} = \frac{y'(t)}{x'(t)}, \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\frac{dy}{dx}}{x'(t)}$$

| input | meaning | condition |
|---|---|---|
| `xt` | x(t) | |
| `yt` | y(t) | |
| `t0` | the parameter value | |

| output | meaning |
|---|---|
| `dy_dx` | dy/dx at t0 |
| `d2y_dx2` | $d^2y/dx^2$ at t0 |

- **Method:** $dy/dx = y$'(t)/x'(t) and d^2y/dx^2 = (d/dt of dy/dx)/x'(t), exact derivatives
- **Assumes:** $dx/dt \neq 0$ at t0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('parametric_slope', xt='t^2', yt='t^3', t0=1)
# -> dy_dx = 1.5, d2y_dx2 = 0.75   (SOLVED)
```

```
calculus_ai999x.solve('parametric_slope', xt='cos(t)', yt='sin(t)', t0=0.7853981633974483)
# -> dy_dx = -1, d2y_dx2 = -2.82843   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find dy/dx for x = t^2, y = t^3 at t = 1. | SOLVED | dy_dx = 1.5, d2y_dx2 = 0.75 |
| Find dy/dx for x = t^2, y = t^3 at t = 0. | INVALID_INPUT | Slope and concavity of a parametric curve: dx/dt = 0 at this parameter: the tangent is vertical (dy/dx undefined). |

## Area in polar coordinates — `polar_area`

The area swept by the polar curve $r = r$(theta) between $\theta = \alpha$ and $\theta = \beta$.

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r(\theta)^2\, d\theta$$

| input | meaning | condition |
|---|---|---|
| `r` | the polar curve r(theta) | |
| `al` | alpha, the start angle (radians) | |
| `be` | beta, the end angle (radians) | |

| output | meaning |
|---|---|
| `area` | the area |

- **Method:** $(1/2)$ integral of $r^2$ by tanh-sinh quadrature
- **Assumes:** $\alpha < \beta \leq \alpha + 2$ pi; r continuous
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('polar_area', r='1 + cos(theta)', al=0, be=6.283185307179586)
# -> area = 4.71239   (SOLVED)
```

```
calculus_ai999x.solve('polar_area', r='2', al=0, be=1.5707963267948966)
# -> area = 3.14159   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the area enclosed by the polar curve r = 1 + cos(theta) from theta = 0 to theta = 2pi. | SOLVED | area = 4.71239 |
| Find the area of the polar region r = 2 from theta = 0 to theta = pi/2. | SOLVED | area = 3.14159 |
| Find the area enclosed by the polar curve r = 1 + cos(theta) from theta = 0 to theta = 4pi. | INVALID_INPUT | Area in polar coordinates: the angle interval may span at most one full turn (2 pi). |

## Gamma function — `gamma_function`

The value of the gamma function (Gamma(n) = (n-1)! for whole n).

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\,dt$$

| input | meaning | condition |
|---|---|---|
| `x` | the argument x | |

| output | meaning |
|---|---|
| `gamma` | Gamma(x) |

- **Method:** the standard library's Lanczos-type approximation (double precision)
- **Assumes:** x is not 0 or a negative whole number; $x < 171.6$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('gamma_function', x=5)
# -> gamma = 24   (SOLVED)
```

```
calculus_ai999x.solve('gamma_function', x=0.5)
# -> gamma = 1.77245   (SOLVED)
```

```
calculus_ai999x.solve('gamma_function', x=4.5)
# -> gamma = 11.6317   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the gamma function at x = 4.5. | SOLVED | gamma = 11.6317 |
| Find Gamma(5). | SOLVED | gamma = 24 |
| Find Gamma(-2). | INVALID_INPUT | Gamma function: the gamma function is undefined at 0 and negative whole numbers, and overflows beyond x = 171. |

### Beta function — `beta_function`

The beta function B(a, b) for positive a and b.

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

| input | meaning | condition |
|---|---|---|
| `a` | the first argument (> 0) | |
| `b` | the second argument (> 0) | |

| output | meaning |
|---|---|
| `beta` | B(a, b) |

- **Method:** exp(ln Gamma(a) + ln Gamma(b) - ln Gamma(a + b))
- **Assumes:** $a > 0$; $b > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('beta_function', a=2, b=3)
# -> beta = 0.0833333   (SOLVED)
```

```
calculus_ai999x.solve('beta_function', a=0.5, b=0.5)
# -> beta = 3.14159   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the beta function B(2, 3). | SOLVED | beta = 0.0833333 |
| Find the beta function B(0.5, 0.5). | SOLVED | beta = 3.14159 |
| Evaluate the beta function B(-1, 3). | INVALID_INPUT | Beta function: this capability needs a > 0 and b > 0. |

### Error function — `error_function`

The error function erf(x) and the complementary error function erfc(x) = 1 - erf(x).

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$$

| input | meaning | condition |
|---|---|---|
| `x` | the argument x | |

| output | meaning |
|---|---|
| `erf` | erf(x) |
| `erfc` | erfc(x) |

- **Method:** the standard library's erf and erfc (double precision)

- **Assumes:** x real
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('error_function', x=0)
# -> erf = 0, erfc = 1   (SOLVED)
```

```
calculus_ai999x.solve('error_function', x=0.5)
# -> erf = 0.5205, erfc = 0.4795   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate erf(0.5). | SOLVED | erf = 0.5205, erfc = 0.4795 |
| Find the error function at x = 0. | SOLVED | erf = 0, erfc = 1 |

## Regularized incomplete gamma functions — `regularized_gamma`

The regularized lower and upper incomplete gamma functions P(a, x) and Q(a, x) = 1 - P(a, x).

$$P(a,x) = \frac{1}{\Gamma(a)} \int_0^x t^{a-1} e^{-t}\, dt$$

| input | meaning | condition |
|---|---|---|
| `a` | the shape a (> 0) | |
| `x` | the upper limit x (>= 0) | |

| output | meaning |
|---|---|
| `P` | P(a, x) = gamma(a, x)/Gamma(a) |
| `Q` | Q(a, x) = Gamma(a, x)/Gamma(a) |

- **Method:** power series for $x < a + 1$, Lentz continued fraction otherwise (the smaller of P and Q computed directly)
- **Assumes:** $a > 0$; $x \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('regularized_gamma', a=1, x=2)
# -> P = 0.864665, Q = 0.135335   (SOLVED)
```

```
calculus_ai999x.solve('regularized_gamma', a=2, x=3)
# -> P = 0.800852, Q = 0.199148   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the regularized incomplete gamma function P(2, 3). | SOLVED | P = 0.800852, Q = 0.199148 |

| question | outcome | result |
|---|---|---|
| Find the regularized incomplete gamma function P(-1, 3). | INVALID_INPUT | Regularized incomplete gamma functions: this capability needs a > 0 and x >= 0. |

## Bessel function of the first kind — `bessel_j`

The Bessel function J_n(x) of whole-number order n.

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,(m+n)!} \left(\frac{x}{2}\right)^{2m+n}$$

| input | meaning | condition |
|---|---|---|
| `n` | the order n | |
| `x` | the argument x | |

| output | meaning |
|---|---|
| `J` | J_n(x) |

- **Method:** hypergeometric series at 30 digits
- **Assumes:** $0 \le n \le 50$; |x| <= 1000
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('bessel_j', n=0, x=0)
# -> J = 1   (SOLVED)
```

```
calculus_ai999x.solve('bessel_j', n=0, x=1)
# -> J = 0.765198   (SOLVED)
```

```
calculus_ai999x.solve('bessel_j', n=1, x=1)
# -> J = 0.440051   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Evaluate the Bessel function J_0(1). | SOLVED | J = 0.765198 |
| Evaluate the Bessel function of the first kind of order 1 at x = 1. | SOLVED | J = 0.440051 |

## Riemann zeta function — `riemann_zeta`

The Riemann zeta function zeta(s) = sum $1/n^s$ for $s > 1$ (the p-series sum).

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

| input | meaning | condition |
|---|---|---|
| `s` | the argument s (> 1) | |

| output | meaning |
|---|---|
| `zeta` | zeta(s) |

- **Method:** Euler-Maclaurin / Borwein evaluation at 30 digits
- **Assumes:** $s > 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('riemann_zeta', s=2)
# -> zeta = 1.64493   (SOLVED)
```

```
calculus_ai999x.solve('riemann_zeta', s=4)
# -> zeta = 1.08232   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find zeta(2). | SOLVED | zeta = 1.64493 |
| Evaluate the Riemann zeta function at s = 4. | SOLVED | zeta = 1.08232 |
| Find zeta(1). | INVALID_INPUT | Riemann zeta function: this capability covers s > 1 (where the defining series converges). |

## Legendre polynomial — `legendre_polynomial`

The value of the Legendre polynomial P_n(x).

$$(k+1)P_{k+1}(x) = (2k+1)\,x\,P_k(x) - k\,P_{k-1}(x)$$

| input | meaning | condition |
|---|---|---|
| `n` | the degree n | |
| `x` | the argument x | |

| output | meaning |
|---|---|
| `P` | P_n(x) |

- **Method:** Bonnet's three-term recurrence $(k+1)$ P_(k+1) = (2k+1) x P_k - k P_(k-1)
- **Assumes:** $0 \le n \le 200$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('legendre_polynomial', n=2, x=0.5)
# -> P = -0.125   (SOLVED)
```

```
calculus_ai999x.solve('legendre_polynomial', n=3, x=0.5)
# -> P = -0.4375   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the Legendre polynomial P_3(0.5). | SOLVED | P = -0.4375 |
| Find the Legendre polynomial of degree 2 at x = 0.5. | SOLVED | P = -0.125 |

## Chebyshev polynomial — `chebyshev_polynomial`

The value of the Chebyshev polynomial of the first kind T_n(x) (T_n(cos t) = cos(n t)).

$$T_{k+1}(x) = 2x\,T_k(x) - T_{k-1}(x)$$

| input | meaning | condition |
|---|---|---|
| `n` | the degree n | |
| `x` | the argument x | |

| output | meaning |
|---|---|
| `T` | T_n(x) |

- **Method:** the recurrence T_(k+1) = 2x T_k - T_(k-1)
- **Assumes:** $0 \le n \le 500$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('chebyshev_polynomial', n=3, x=0.5)
# -> T = -1   (SOLVED)
```

```
calculus_ai999x.solve('chebyshev_polynomial', n=2, x=0.3)
# -> T = -0.82   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate the Chebyshev polynomial T_3(0.5). | SOLVED | T = -1 |
| Find the Chebyshev polynomial of degree 2 at x = 0.3. | SOLVED | T = -0.82 |

## Fourier coefficients — `fourier_coefficients`

The Fourier coefficients a_0, a_k and b_k of f on [-L, L] (series a_0/2 + sum a_k cos(k pi x/L) + b_k sin(k pi x/L)).

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{k\pi x}{L}\,dx, \quad b_k = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{k\pi x}{L}\,dx$$

| input | meaning | condition |
|---|---|---|
| `f` | the function f(x) | |
| `L` | the half-period L (> 0) | |
| `k` | the order k | |

| output | meaning |
|---|---|
| `a0` | $a_0 = (1/L)$ integral of f |
| `a_k` | a_k |
| `b_k` | b_k |

- **Method:** tanh-sinh quadrature on the k half-wave subintervals at 20 digits
- **Assumes:** $L > 0$; f continuous on [-L, L]; $1 \le k \le 50$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
calculus_ai999x.solve('fourier_coefficients', f='x', L=3.141592653589793, k=2)
# -> a0 = 0, a_k = 0, b_k = -1   (SOLVED)
```

```
calculus_ai999x.solve('fourier_coefficients', f='x^2', L=3.141592653589793, k=1)
# -> a0 = 6.57974, a_k = -4, b_k = 0   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the Fourier coefficients of order 2 of f(x) = x on [-pi, pi]. | SOLVED | a0 = 0, a_k = 0, b_k = -1 |
| Find the Fourier coefficients of order 1 of x^2 on the interval [-pi, pi]. | SOLVED | a0 = 6.57974, a_k = -4, b_k = 0 |
| Find the Fourier coefficients of order 1 of 1/x on [-1, 1]. | INVALID_INPUT | Fourier coefficients: the function is not continuous (or not real) on the whole interval [-1, 1]. |